



**COMSATS University Islamabad,
Sahiwal Pakistan**

Skill Forge

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COMSATS Institute of Information Technology, Islamabad,
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**In partial fulfillment
of the requirement for the degree of**

***Bachelor of Science in Software Engineering (2022-
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Executive Summary

Skill Forge is an innovative e-learning platform designed to make quality education accessible, interactive, and engaging for learners of all ages and backgrounds. Unlike traditional learning management systems, Skill Forge is tailored to provide a dynamic and intuitive experience, offering a wide variety of courses in fields such as technology, business, arts, and languages.

The platform allows students to browse and enroll in courses, track their learning progress, participate in discussion forums, and communicate in real-time with instructors through integrated chat features. This direct interaction enhances the learning experience, making it more collaborative and engaging.

With its user-friendly interface, Skill Forge ensures that users from all technical backgrounds can easily navigate the application. For educators, Skill Forge offers tools for content management, student assessment, and real-time feedback, creating a comprehensive and supportive learning environment.

The app's development will follow an agile methodology, allowing for the incremental implementation of features based on functional requirements. Non-functional requirements such as usability, performance, reliability, security, and scalability will be prioritized to deliver a seamless user experience.

The testing and evaluation phase will include unit testing, integration testing, system testing, and user acceptance testing to ensure that all functionalities operate as intended and meet user expectations.

In summary, Skill Forge offers a dedicated and modern space for interactive and productive learning, enhancing knowledge sharing and skill development, and supporting students and professionals in achieving their educational and career goals.

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Abbreviations

UC	Use Case
API	Application Programming Interface
JS	JavaScript
SRS	Software Require Specification
SDD	Software Design Description
HCI	Human-Computer Interaction
SDLC	Software Development Life Cycle
DBMS	Database Management System
UT	Unit Testing
FT	Functional Testing
MVC	Model-View-Controller
GUI	Graphical User Interface

IDEs	Integrated development environments
CSS	Cascading Style Sheet
NoSQL	Not Only SQL (Flexible Database Format)
CRUD	Create, Read, Update, Delete

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Chapter 1

Introduction

1. Introduction

In today's fast-paced digital world, the demand for accessible, high-quality education is greater than ever. Traditional methods of education are rapidly evolving to meet the needs of learners who require flexibility, interactivity, and continuous access to learning resources.

Skill Forge is an innovative e-learning platform developed to address these emerging educational needs by offering a dynamic and intuitive learning environment for students and professionals.

Skill Forge connects learners and educators through an interactive platform that offers a wide range of courses across multiple disciplines, including technology, business, arts, and languages. The platform is designed to be user-friendly and accessible to individuals from all technical backgrounds, ensuring that learning is inclusive, engaging, and convenient for all users.

The core vision of Skill Forge is to empower users by providing features such as course enrollment, real-time progress tracking, road-map generation, RAG based assistant, direct communication with instructors through video call platform, and opportunities to give feedback and review courses. These features collectively create a supportive and collaborative learning experience.

To ensure a high standard of quality, Skill Forge emphasizes not only functional aspects but also non-functional requirements such as usability, performance, security, portability, reliability, and scalability. The platform is mobile-optimized and developed using Android Studio for the frontend and Firebase for the backend, ensuring a secure, responsive, and real-time experience for users.

The development of Skill Forge follows an agile methodology, allowing iterative improvements based on user feedback and evolving needs. The platform's design also considers essential operational assumptions, such as reliable internet connectivity and device compatibility, ensuring that it remains a reliable tool for learners anywhere, anytime.

This report provides a detailed overview of the Skill Forge project, covering its objectives, design principles, functional and non-functional requirements, use case analysis, and development approach. It reflects the commitment to creating a comprehensive and efficient e-learning platform that enhances educational opportunities and supports lifelong learning in a digital age.

1.1 Brief Overview

The Skill Forge project aims to develop a cutting-edge e-learning mobile application tailored to meet the growing demand for flexible, high-quality education among students and professionals. With a strong focus on mobility, user experience, and interactivity, Skill Forge is designed to make learning more accessible, engaging, and effective for users from all walks of life.

This mobile application will primarily be developed for Android devices using Android Studio as the development environment. For backend operations, the app will integrate Firebase, which will provide real-time database management, secure user authentication, cloud storage, and smooth data synchronization across devices.

This powerful tech stack ensures that users experience fast performance, seamless navigation, and real-time updates throughout their learning journey. The development of Skill Forge will follow the agile software development methodology, which allows for the flexible and iterative creation of features. Agile encourages active collaboration, continuous testing, and ongoing user feedback. This approach ensures that the platform evolves efficiently with improvements added at each development cycle, making it responsive to user needs and market changes.

Unlike many generic learning platforms, Skill Forge is a dedicated and specialized e-learning solution, initially focusing on a wide variety of technology-related courses. These may include areas such as programming, web development, data science, cybersecurity, and more. In the future, the platform can be expanded to include additional categories based on user interest and demand.

Skill Forge is built with a learner-first mindset, offering a variety of essential features aimed at delivering a complete educational experience. These include:

- Course enrollment and management
- Progress tracking and personalized dashboards
- Real-time communication
- Interactive quizzes and assignments
- Profile Management
- Course rating and feedback systems
- Personalized assistant for each course
- Roadmap generation

These features are designed to support active learning, encourage collaboration, and help users stay motivated and accountable throughout their learning process.

The platform's mobile-friendly and responsive design ensures users can access learning materials on the go, whether from home, the office, or while commuting. Its user-centric interface focuses on simplicity, clarity, and ease of use, making it suitable for learners of all technical backgrounds from beginners to experts.

By combining functionality, design, and real-time interaction, Skill Forge positions itself as a powerful educational tool in the modern digital learning space. The project reflects a strong commitment to improving learning experiences and empowering users through technology.

1.2 Relevance to Course Modules

The development of Skill Forge is not just about building a mobile application; it's a comprehensive demonstration of the practical application of various software engineering concepts and technical skills gained throughout the course.

This project effectively bridges academic learning with real-world application, ensuring the delivery of a high-quality, modern e-learning platform.

1.2.1 Technical Skills

Skill Forge makes extensive use of technical knowledge in mobile application development. The front-end is built using Android Studio, leveraging Kotlin and XML for crafting a responsive and functional user interface.

On the backend, Firebase is used for handling real-time data, user authentication, and cloud storage. Skills acquired in programming, mobile UI development, and integration of third-party services are applied throughout the app's development.

Features such as login functionality, course lists, real-time chat, and data synchronization are developed using tools and techniques learned during the software engineering course, making Skill Forge a strong demonstration of practical programming expertise.

1.2.2 Software Engineering Principles

The project adheres closely to the principles and methodologies taught in software engineering. Starting with structured requirement gathering and analysis, every feature of the app was planned with clarity and precision.

Using agile methodology, the development was broken into manageable sprints, allowing for incremental updates and continuous improvement based on feedback. Design patterns and clean coding practices were followed to maintain code readability and reusability.

Regular testing and debugging at each phase ensured quality assurance, while maintenance considerations were incorporated from the beginning to allow future scalability and updates.

1.2.3 Database Design

Skill Forge leverages both Firebase Realtime Database and Cloud Firestore to manage backend operations. These databases store user profiles, course materials, progress records, chat messages, and

user feedback in an organized, structured way.

Principles of data normalization, data security, and efficient query handling are applied to ensure smooth, real-time access and manipulation of data. The integration also includes rules for user data privacy, allowing safe and authenticated access to sensitive information, which aligns with industry standards in cloud-based database design.

1.2.4 System Architecture

The architecture of Skill Forge follows a modular microservices, where responsibilities are clearly separated across different layers of the system. The client-side (mobile app) handles user interactions, while Firebase handles data operations on the server side. This division ensures that each component can be developed and updated independently.

The use of MVC (Model-View-Controller) patterns, modular course components, and secure authentication flows helps maintain a clean architecture. Real-time updates, like course progress and chat, are supported through Firebase's built-in features, enabling an efficient and responsive architecture suitable for high-volume user activity.

1.2.5 UI/UX Design

Creating a smooth and engaging learning experience requires strong attention to UI/UX design principles. Drawing from concepts studied in Human-Computer Interaction (HCI), the user interface of Skill Forge is built to be clean, intuitive, and mobile-friendly. Elements like readable fonts, appropriate spacing, color contrast, and consistent navigation patterns make the app accessible to users of all ages and technical backgrounds.

Feedback loops such as progress indicators, confirmation messages, and alerts further enhance user interaction. Special care has been taken to make the experience both aesthetically pleasing and functionally efficient, reducing cognitive load and improving overall satisfaction.

1.2.6 Project Management

The Skill Forge project is executed with strong project management discipline. It began with clear goal setting, timeline estimation, and division of tasks. Sprint planning was used to break development into phases, with regular reviews and retrospectives to track progress and address challenges.

Tools like Trello or Jira may be used (or simulated) to monitor task completion and ensure on-time delivery. This method of working promotes accountability, smooth collaboration, and timely execution. Risk management and contingency planning were also considered to reduce potential delays and ensure a successful launch.

1.2.7 Networking

Real-time features such as live chat, quiz submissions, and progress updates rely heavily on networking concepts. Skill Forge ensures smooth client-server communication using Firebase's real-time capabilities over reliable internet protocols. Concepts such as data synchronization, latency management, and event-based communication are employed to ensure a responsive user experience.

For example, when a user submits a quiz or sends a message to the instructor, the data is instantly uploaded and visible to others in real time. The implementation reflects a clear understanding of network architecture, protocols, and bandwidth optimization techniques.

1.3 Project Background

The need for the Skill Forge project arises from various challenges observed in the current education system, particularly with the growing shift toward remote and flexible learning environments. These challenges affect both learners and educators, revealing gaps in accessibility, interactivity, and efficiency in existing e-learning solutions.

Before Skill Forge, learners frequently encountered difficulties in accessing high-quality educational content on a single, unified platform. Many traditional e-learning systems were either limited in functionality or failed to offer an engaging and user-centric learning experience.

While some platforms focused heavily on content delivery, they lacked real-time interaction features such as live chat, feedback systems, or collaborative discussion. Others were not well-optimized for mobile devices, making it difficult for users to learn on-the-go a growing necessity in today's fast-paced lifestyle.

This lack of interactivity and flexibility made it harder for students and working professionals to stay engaged with their studies. Learners often had to juggle between multiple apps or platforms to access materials, communicate with instructors, submit assignments, and track their own progress. This disjointed experience reduced learning effectiveness and user satisfaction.

On the other hand, educators also faced significant hurdles. Managing course content, organizing quizzes and assignments, tracking learner performance, and communicating with students across multiple tools proved time-consuming and inefficient. The absence of an integrated system made it challenging to maintain a structured, smooth, and interactive teaching environment.

Furthermore, during recent years particularly during and after the COVID-19 pandemic the importance of robust online education platforms has become more evident than ever. Educational

institutions, instructors, and independent learners all needed a solution that is not just functional but intuitive, responsive, secure, and adaptable to various learning styles and environments.

Skill Forge was conceptualized to solve these problems by combining the core features of effective e-learning into a single, mobile-optimized application.

It is designed to bridge the gap between learners and educators by offering real-time communication tools, a centralized content management system, progress tracking, and interactive learning features all in one place. The platform ensures that learners are not just passive consumers of information but active participants in a collaborative and flexible learning journey.

This project aims to redefine digital learning by providing an inclusive, modern, and accessible solution that supports both educational goals and user convenience, setting a new standard for mobile e-learning applications.

1.4 Methodology and Software Lifecycle

Skill Forge is a comprehensive software project with diverse and evolving requirements that reflect the dynamic nature of the education sector. In order to manage the complexity and changing scope effectively, it is essential to adopt a development methodology and software lifecycle model that encourages flexibility, teamwork, and continuous improvement throughout the project.



Figure 1.1 Agile Methodology

To meet these needs, the Agile methodology has been selected for the Skill Forge project. Agile is well-known for its iterative approach, where the development process is broken down into small, manageable increments called sprints.

Each sprint delivers a working piece of the application, allowing the team to regularly review progress,

gather user feedback, and make necessary adjustments. This continuous loop of planning, development, testing, and feedback ensures that the final product is closely aligned with user expectations and educational goals.

Agile promotes close collaboration among developers, designers, testers, and stakeholders, enabling a shared understanding of the project vision and faster decision-making. For Skill Forge, this means that features such as course enrollment, live chat, quizzes, progress tracking, and video call platform can be developed in phases, tested early, and refined quickly based on real-time feedback from users and instructors.

Moreover, the Software Development Life Cycle (SDLC) followed in this project includes all critical phases from requirement gathering, analysis, and design to implementation, testing, deployment, and maintenance. In each phase, Agile principles guide the team to adapt and evolve the solution efficiently.

By using Agile, Skill Forge benefits from

- Early and continuous delivery of valuable features.
- High adaptability to changing user needs and feedback.
- Improved communication across team members and with stakeholders.
- Regular testing and integration, ensuring a stable and reliable product.
- Faster time-to-market for core features.

Additionally, tools like version control (Git), project management boards (such as Trello or Jira), and real-time collaboration platforms are used to support Agile practices and ensure smooth coordination among the development team.

In conclusion, the Agile methodology, combined with a well-structured software lifecycle model, ensures that Skill Forge is developed in a professional, flexible, and user-centric manner. It allows the team to deliver a high-quality, scalable, and future-ready mobile e-learning application that continuously evolves with user needs and technological advancements.

1.5 Rationale behind Selected Agile Model

Agile is a modern and widely adopted software development methodology that emphasizes adaptability, collaborative teamwork, and continuous improvement. It is particularly effective for complex and user-driven projects like Skill Forge, where user needs and technology trends evolve rapidly. By applying Agile principles, the development process becomes more flexible, transparent,

and focused on delivering maximum value to users at every stage.

Agile allows for constant engagement with end-users students, instructors, and administrators ensuring their voices are heard and integrated into the application's evolution. This collaborative and feedback-driven approach helps create an app that is not only technically sound but also meaningful and useful to its target audience.[\[2\]](#)

Agile methodology for Skill Forge is grounded in the following key principles

- **Iterative Development**

Skill Forge will be developed in short development cycles called iterations or sprints. Each iteration lasts a few weeks and focuses on delivering specific features or improvements. This approach allows the team to build, test, and improve the app in manageable phases, ensuring steady progress and allowing for early detection of issues.

For example, one iteration may focus on creating the login and registration system, while the next may develop the course enrollment module. At the end of each cycle, the team reviews what was achieved and plans the next steps, allowing constant refinement and enhancement of the platform based on feedback and performance.

- **Incremental Delivery**

Instead of delivering the entire application at once, Skill Forge will be released incrementally, with each release adding meaningful functionality. This ensures that users start benefiting from the app early in the development cycle, even while new features are still being built.

For instance, an initial version may include core features like account creation and course listing. Subsequent releases may then add chat support, quizzes, progress tracking, or discussion forums. This staged delivery approach provides early value, reduces risk, and allows for gradual scaling based on user feedback.

- **Continuous Feedback**

One of the strongest pillars of Agile is regular and real-time feedback. Skill Forge will actively collect input from its users through in-app feedback forms, surveys, pilot testing, and direct communication with instructors and learners. This ensures that any bugs, usability issues, or feature requests are identified and addressed promptly.

By involving users throughout the process, the development team can prioritize features that matter most to the audience and eliminate unnecessary or confusing elements. Feedback-driven development ensures the app remains intuitive, relevant, and impactful.

- **Adaptive Planning**

Skill Forge's development plan is not fixed or rigid. Instead, it follows adaptive planning, where the

roadmap is continuously revised based on real-world inputs, user needs, and emerging trends in e-learning and mobile technology.

If a new need arises such as adding AI-based course recommendations or offline learning modes the development plan can be adjusted to include these new priorities. This flexibility makes the platform future-ready and able to respond to changes in educational demands, ensuring long-term relevance and user satisfaction.

Chapter 2

Problem Definition

2. Problem Definition

This chapter discusses the specific problem that the Skill Forge project aims to solve, along with the expected outcomes resulting from its development.

2.1 Problem Statement

The Skill Forge project seeks to address a pressing and growing challenge in today's educational landscape the limited availability of comprehensive, interactive, and mobile-optimized e-learning solutions that effectively serve both students and working professionals for learning and earning both perspectives.

Despite a global shift toward digital learning, most existing platforms either lack essential features or are fragmented in their approach, leading to inconsistent and often frustrating user experiences.

Many current e-learning systems fall short in delivering a truly integrated environment one that combines real-time communication with instructors, intuitive course navigation, personalized learning pathways, and community engagement all within a seamless mobile application.

Learners often find themselves switching between multiple tools for classes, communication, assessments, and progress tracking, which disrupts the flow of learning and affects motivation and performance.

Furthermore, accessibility and usability remain significant concerns. Platforms not designed with a mobile-first mindset tend to alienate users who rely on smartphones as their primary device, especially in regions where access to laptops or high-speed internet is limited. This digital divide creates barriers for learners from diverse backgrounds who are eager to pursue education in a more convenient, flexible manner.

From the educator's perspective, challenges include managing course content efficiently, tracking learner progress, responding to queries in real time, and gathering feedback in a streamlined manner. Most available systems lack built-in tools that facilitate this level of engagement and oversight in a user-friendly way.

Skill Forge aims to bridge these critical gaps by offering a unified, mobile-centric platform that empowers both learners and educators. The app will enable users to explore and enroll in courses, monitor their academic journey, participate in quizzes, engage in live discussions, communicate with instructors, and submit feedback all through a clean and accessible interface.

In summary, Skill Forge addresses the shortcomings of traditional and existing e-learning platforms by delivering a feature-rich, scalable, and student-centered experience designed to meet the modern

demands of digital education. It seeks not only to improve access to quality learning but to redefine how education is delivered in a fast-paced, mobile-driven world.

2.2 Deliverables and Development Requirements

The Skill Forge project is structured around a series of well-defined deliverables and development requirements that ensure a comprehensive, scalable, and user-centric learning platform. Each component has been carefully designed to support the learning process, foster collaboration, and empower users both students and educators. The key deliverables include

2.2.1 COURSE ENROLLMENT

The platform will feature an extensive course catalog that users can easily browse by categories, difficulty level, instructor, or popularity. Each course listing will include detailed information such as course description, syllabus, duration, instructor bio, course ratings, and pricing options.

Both free and premium courses will be available, with a seamless payment gateway integrated for paid courses. The enrollment process will be intuitive, guiding users step-by-step with prompts and confirmations. Enrolled courses will be automatically added to the user's dashboard for easy access.

2.2.2 Progress Tracking

To keep users engaged and goal-oriented, Skill Forge will implement a robust progress tracking system. Each course will display real-time tracking of lessons completed, quiz scores, and assignment submissions. The user dashboard will provide a visual overview (e.g., progress bars, badges, and milestone notifications) to encourage learning continuity. Additionally, users will receive performance analytics and personalized recommendations based on their activity and progress.

This system will motivate users to stay on track and complete their courses efficiently. Regular updates and reminders will further support consistent learning habits and user retention.

2.2.3 Real-Time Communication

Skill Forge will integrate real-time messaging functionalities such as live Q&A sessions with instructors, include voice and video calling support for virtual office hours or group meetings.

Notifications will be sent for new messages, announcements, or session reminders to keep communication timely and efficient. These real-time features will promote immediate feedback, personal engagement, and an interactive learning atmosphere.

2.2.4 Quizzes and Assessments

Courses will include dynamic quizzes, timed tests, and assignments tailored to specific learning objectives. The system will support multiple question types (MCQs, short answers, coding challenges, etc.) with automatic grading where applicable.

Learners will receive instant feedback, and performance analytics will be stored for instructors to review. Advanced settings will allow for retakes, custom scoring, and difficulty levels to align with various learner profiles.

2.2.5 Feedback and Reviews

After completing a course or module, learners can rate the content and leave constructive feedback. Reviews will highlight strengths and suggest improvements, helping future users choose relevant courses. Instructors can also respond to feedback and update course materials accordingly. This review loop fosters accountability and ensures ongoing content refinement.

2.2.6 Administrative Tools

Admins will have access to a secure backend via the app to manage user data, upload new content, approve course listings, and monitor system activity. Analytics dashboards will offer insights into course engagement, user retention, and system performance. Admin tools will include features like user role management (e.g., student, instructor, moderator), reporting and analytics generation, content scheduling, and more

2.2.7 Regular Updates and Maintenance

Post-launch, Skill Forge will undergo regular version releases to introduce new features, fix bugs, and maintain compatibility with evolving Android OS versions and Firebase updates. A roadmap for upcoming features will be shared publicly to keep the community engaged. Automated monitoring tools will flag system issues, and regular maintenance windows will be scheduled to keep services uninterrupted.

These deliverables and development requirements collectively contribute to the successful creation and continued improvement of Skill Forge, aligning with the project's objectives of enhancing communication, collaboration, and community engagement among university students.

2.3 Current System

Currently, the educational environment lacks a centralized, integrated e-learning platform that effectively connects students, instructors, and administrators within a single digital ecosystem. This gap results in fragmented communication channels, inconsistent access to learning materials, and an inefficient system for managing academic activities.

Course content is often distributed across multiple uncoordinated platforms such as WhatsApp, Google Drive, or standalone learning apps, making it difficult for students to locate essential resources.

Assignments, quizzes, and announcements are shared in a scattered manner, causing confusion and

missed deadlines. Without a unified platform, instructors face challenges in organizing and delivering content, while students struggle to maintain consistent engagement and track their learning progress effectively.

Moreover, student-instructor interaction is minimal and often delayed due to the absence of realtime communication tools. This not only affects the learning experience but also impacts timely feedback and mentoring, which are essential components of academic success. Peer-to-peer collaboration is also limited, reducing opportunities for group discussions, cooperative projects, and community-based learning.

Administratively, managing course enrollments, student records, academic performance, and feedback is cumbersome and time-consuming. There is no dedicated system for tracking student progress, monitoring course effectiveness, or analyzing engagement metrics. As a result, institutions face difficulties in making data-driven decisions for academic planning and improvement.

The lack of mobile-optimized solutions further widens the accessibility gap, especially for students who rely on smartphones as their primary learning device. Offline access to course materials is generally unavailable, restricting learning opportunities for students in areas with poor or unreliable internet connectivity.

In summary, the current system is

- Disjointed and inefficient in course delivery and management
- Devoid of centralized academic progress tracking tools
- Inaccessible on mobile and underdeveloped for inclusive learning
- Unfit for scalable growth or long-term academic digital transformation

To overcome these challenges and modernize the educational experience, the development of Skill Forge is underway. This platform aims to unify all key learning components under one roof . Skill Forge will not only bridge the current digital divide but also lay the foundation for a future-ready, student-centered academic environment.

Chapter 3

Requirement Analysis

3. Requirement Analysis

3.1 Use Case Diagrams

A use case diagram is a type of behavioral diagram in Unified Modeling Language (UML) used to visually understand the functional requirements of a system. It shows the interactions between actors (which could be users, other systems, or hardware) and the system itself, detailing the different use cases (specific actions or functions) that the system will perform. The goal is to illustrate how users or other systems will interact with the system to achieve a particular goal.

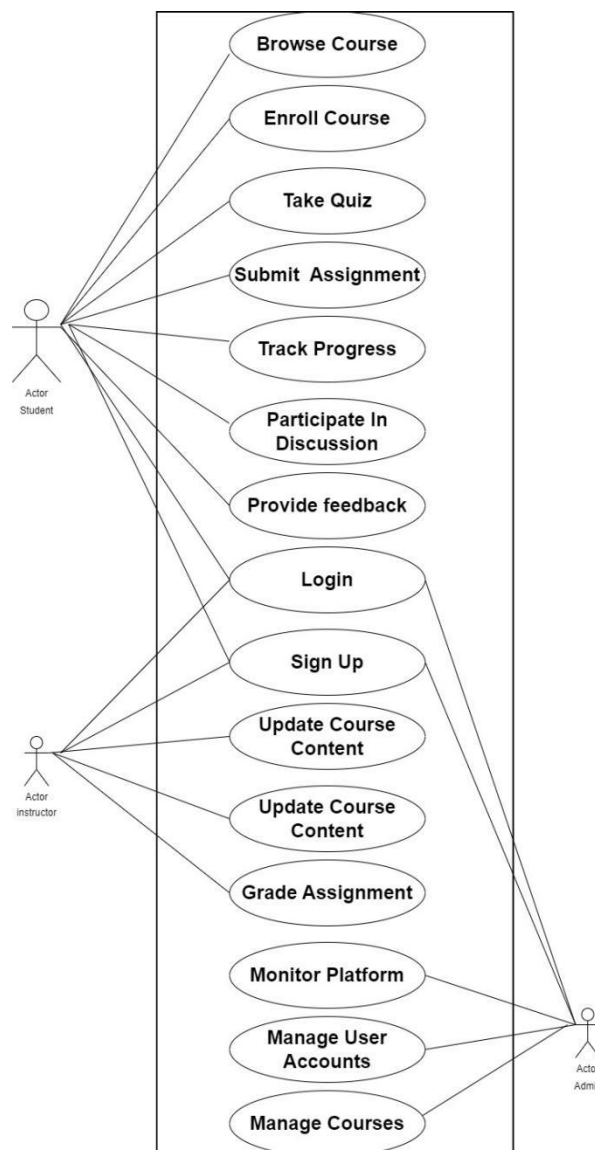


Figure 3.1 Use Case Diagram

3.2 Use Case Descriptions

The table below indicates a comprehensive use case template filled in with an example drawn from the Skill Forge e-learning app and course management system.

The Table 3.1 , **Login/Sign Up** use case defines how a user gains access to the Skill Forge application by either logging into an existing account or creating a new one. This process begins when the user initiates a request to access the system, typically by selecting the login or sign-up option on the application interface. The system then prompts the user to provide the required credentials, such as an email address or phone number along with a password.

For existing users, the system verifies the entered information by matching it against stored records in the database. If the credentials are valid, the user is successfully authenticated and granted access to the application's features. In the case of new users, the sign-up process requires entering necessary details to create an account, after which the system stores this information securely and logs the user into the system.

If incorrect or invalid credentials are entered during login, the system displays an appropriate error message and allows the user to re-enter the details. This ensures both security and usability by preventing unauthorized access while guiding the user to correct mistakes. The entire process depends on an active network connection to validate and store user data. Overall, this use case ensures a secure, efficient, and user-friendly entry point into the application.

Table 3.1: Login/Signup

Use case id	01
Use case name	Login/Sign Up
Actors	User
Description	A user logs into the application or sign up if they do not have an account.
Trigger	User intends to requests to log in or sign up.
Pre-condition	The user must have an account or be willing to create one.
Postcondition	The actor is now logged into the application or has successfully signed up and logged in.
Normal flow	This use case starts when an actor wants to login into the system. The system requests that the actor enter e-mail/phone no. andpassword.
Alternative flow	For example, a Valid e-mail/Phone number or an Invalid Password If in the basic flow the system cannot find the valid e-mail or the password is invalid, an error message is displayed.
Exceptions	Incorrect credentials. Request to re-enter the login details.
Business rules	The system must be connected to the network.

The Table 3.2 **Enroll Course** use case describes how a student registers for a course within the Skill Forge application. This process begins when a student, already logged into their account, selects a course they wish to enroll in. The system then takes this request as a trigger to initiate the enrollment process and ensures that the student meets all necessary conditions before proceeding further.

Once the course is selected, the system checks whether the student satisfies the required prerequisites, such as prior courses or eligibility criteria. It also verifies the availability of seats in the selected course. If all conditions are met, the system prompts the student to confirm their enrollment. Upon confirmation, the system successfully registers the student in the course and updates their learning dashboard accordingly.

In cases where the prerequisites are not fulfilled, the system prevents enrollment and displays an appropriate error message to inform the student. Similarly, exceptions such as a full course or network issues may interrupt the process, requiring the student to try again later. This use case ensures that course enrollment is handled in a structured and controlled manner, maintaining academic requirements and system integrity.

Table 3.2: Enroll Course

Use case id	02
Use case name	Enroll Course
Actors	Students
Description	A student enrolls in a selected course.
Trigger	Student intends to selects the option to enroll in a course.
Precondition	The student must be logged into their account and have the necessary prerequisites.
Post condition	The student is successfully enrolled in the selected course.
Normal flow	The student selects a course to enroll in. The system checks the prerequisites and availability. The student confirms the enrollment.
Alternative flow	If prerequisites are not met, the system displays an error message.
Exceptions	Course is full and network connection issues.
Business Rules	The system must verify prerequisites before enrolling.

The Table 3.3 **Take Quiz** use case explains how a student attempts and submits a quiz within the Skill Forge application. This process begins when a student, who is already enrolled in a course, selects the quiz option from the course page. The system recognizes this action as a trigger and prepares the quiz interface for the student.

Once initiated, the system displays the quiz questions along with any relevant instructions and time limits. The student then answers the questions within the given time frame and submits the quiz upon completion. After submission, the system processes the responses and records the results in the database, making them available for both the student and instructor to review.

If the student fails to submit the quiz within the allotted time, the system automatically submits the quiz to ensure fairness and adherence to rules. In case of issues such as network failure or technical errors during submission, the process may be interrupted, requiring appropriate handling or reattempts. This use case ensures a structured and time-bound assessment process, maintaining consistency and integrity in evaluating student performance.

Table 3.3 Take Quiz

Use case id	03
Use case name	Take Quiz
Actors	Students
Description	A student takes a quiz for a course they are enrolled in.
Trigger	Student intends to select the quiz option from the course page.
Precondition	The student must be enrolled in the course.
Postcondition	The quiz is submitted and results are recorded.
Normal flow	The student selects the quiz to take. The student completes and submits the quiz.
Alternative flow	If the student does not complete the quiz in the allotted time, the system automatically submits it.
Exceptions	Network connection issues. Technical problems with quiz submission.
Business Rule	Quiz must be completed within the time limit.

The Table 3.4 **Submit Assignment** use case describes how a student uploads and submits an assignment within the Skill Forge application. This process begins when a student, who is already enrolled in a course, selects the assignment submission option from the course page. The system treats this action as a trigger and prepares the submission interface for the student.

Once the student selects a specific assignment, the system displays a submission form where the student can upload the required file. The student attaches the assignment file according to the given instructions and submits it through the system. After submission, the system processes the request and provides a confirmation message, ensuring that the assignment has been successfully received and recorded.

If the student attempts to submit the assignment after the deadline, the system prevents the submission and displays an appropriate error message. Additionally, issues such as network failures or file upload errors may interrupt the process, requiring the student to retry. This use case ensures that assignment submissions are handled in a structured and time-bound manner, maintaining academic rules and proper record keeping.

Table 3.4 Assignment

Use case id	04
Use case name	Submit Assignment
Actors	Student
Description	A student submits an assignment for a course.
Trigger	Student intends to select the assignment submission option from the course page.
Precondition	The student must be enrolled in the course and the assignment must be available for submission.
Postcondition	The assignment is submitted and confirmation is received.
Normal flow	The student selects the assignment to submit. The student uploads the assignment file and submits it.
Alternative flow	If the submission deadline has passed, the system displays an error message.
Exceptions	Network connection issues and file upload errors.
Business Rules	Assignments must be submitted before the deadline.

The Table 3.5 **Track Progress** use case explains how a student monitors their learning performance within the Skill Forge application. This process begins when a student, who is already logged into their account, navigates to the progress tracking section of the system. The system recognizes this action as a trigger and retrieves the relevant data associated with the student's enrolled courses.

Once accessed, the system displays a detailed progress report that includes information such as course completion status, quiz scores, assignment submissions, and overall performance. This allows the student to clearly understand their strengths, weaknesses, and level of completion in each course. The data is presented in an organized manner to help students easily interpret their learning progress.

The system ensures that all progress data is updated in real-time so that students always view the most accurate and current information. In case of network issues, the progress report may not load properly, requiring the user to retry. This use case provides students with continuous feedback on their performance, helping them stay informed and motivated throughout their learning journey.

Table 3.5 Track Progress

Use case id	05
Use case name	Track Progress
Actors	Student
Description	A student tracks their progress in the courses they are enrolled in.
Trigger	Student intends to navigates to the progress tracking section.
Pre-condition	The student must be logged into their account.
Postcondition	The student views their progress report.
Normal flow	The student navigates to the "Track Progress" section. The system displays the student's progress in each course.
Alternative flow	None
Exceptions	Network connection issues
Business Rules	Progress data must be updated in real-time.

The Table 3.6 **Provide Feedback** use case describes how a student shares their opinions and suggestions regarding a course or the overall Skill Forge platform. This process begins when a student, who is already logged into their account, selects the feedback option from the application. The system treats this action as a trigger and opens the feedback interface for the user.

Once initiated, the system displays a feedback form where the student can enter their comments, ratings, or suggestions. The student fills out the form based on their experience and submits it through the system. After submission, the system processes the input and provides a confirmation message to ensure that the feedback has been successfully recorded.

Table 3.6 Feedback

Use case id	06
Use case name	Provide Feedback
Actors	Student
Description	A student provides feedback for a course or the platform.
Trigger	Student intends to select the feedback option.
Pre-condition	The student must be logged into their account.
Postcondition	The feedback is submitted successfully.
Normal flow	The student selects the feedback option. The system displays the feedback form. The student fills out the form and submits it. The system confirms the submission.
Alternative flow	None
Exceptions	Network connection issues.
Business Rules	Feedback must be stored securely and used for platform improvement.

The Table 3.7 **Update Course Content** use case describes how an instructor modifies and manages course materials within the Skill Forge application. This process begins when an instructor, who is already logged into their account, navigates to the course content management section. The system recognizes this action as a trigger and provides access to the instructor's courses.

The instructor then selects a specific course they wish to update. The system displays the existing course content, which may include lectures, notes, videos, or other learning resources. The instructor reviews the material and makes the necessary changes, such as adding new content, editing existing information, or removing outdated resources. Once the updates are completed, the instructor submits the changes through the system.

After submission, the system processes the request and updates the course content accordingly, ensuring that students have access to the latest and most relevant information. Although there is no alternative flow,

network-related issues may interrupt the process. Additionally, the system maintains a log of all content updates for audit purposes, ensuring transparency and accountability in course management.

Table 3.7 Update course content

Use case id	07
Use case name	Update Course Content
Actors	Instructor
Description	An instructor updates the content of a course.
Trigger	Instructor intends to navigates to the course content management section.
Pre-condition	The instructor must be logged into their account.
Postcondition	The course content is updated successfully.
Normal flow	The instructor selects the course to update. The system displays the current course content. The instructor makes the necessary changes and submits them. The system updates the course content.
Alternative flow	None
Exceptions	Network connection issues.
Business Rules	Content updates must be logged for audit purposes.

The Table 3.8 **Grade Assignment** use case describes how an instructor evaluates and assigns grades to student submissions within the Skill Forge application. This process begins when an instructor, who is logged into their account and has pending assignments to review, navigates to the assignment grading section. The system recognizes this action as a trigger and provides access to the list of submitted assignments.

The instructor selects a specific assignment to grade, and the system displays the student's submitted work. The instructor carefully reviews the assignment and then enters the appropriate grade along with feedback or comments to guide the student. Once the evaluation is complete, the instructor submits the grading details through the system.

After submission, the system records the grade and feedback, making them available to the student for review. Although there is no alternative flow, issues such as network connectivity problems may interrupt the process. The system also ensures that grading is completed within the specified deadline, maintaining fairness and consistency in the evaluation process.

Table 3.8 Grade Assignment

Use case id	08
Use case name	Grade Assignment
Actors	Instructor
Description	An instructor grades an assignment submitted by a student.
Trigger	Instructor intends to navigates to the assignment grading section.
Pre-condition	The instructor must be logged into their account and have assignments to grade.
Postcondition	The assignment is graded, and the grade is recorded.
Normal flow	The instructor selects an assignment to grade. The system displays the submitted assignment. The instructor enters the grade and feedback. The system records the grade and feedback.
Alternative flow	None
Exceptions	Network connection issues.
Business Rules	Grading must be completed within the deadline set for the course.

The Table 3.9 **Monitor Platform** use case describes how an admin oversees the overall performance and functionality of the Skill Forge application. This process begins when an admin, who is already logged into their account, navigates to the monitoring dashboard. The system recognizes this action as a trigger and loads the platform’s performance data.

Once accessed, the system displays real-time information related to system performance, user activity, server status, and potential issues. This data is presented in a structured and easy-to-understand format, enabling the admin to quickly analyze the condition of the platform. The admin carefully reviews this information to identify any errors, performance bottlenecks, or unusual activity that may affect the system.

In addition to real-time data, the dashboard also provides summarized metrics and logs that help the admin understand recent system behavior. The admin can monitor active users, ongoing sessions, and system load to ensure smooth operation. If any issue is detected, the admin can take appropriate action such as investigating logs or initiating corrective measures. The system ensures that all monitoring data is consistently updated, allowing continuous oversight without interruption. This process supports system reliability, helps maintain performance standards, and ensures a stable experience for all users on the platform.

Table 3.9 Monitor Platform

Use case id	09
Use case name	Monitor platform
Actors	Admin
Description	An admin monitors the overall platform for performance and issues.
Trigger	Admin intends to navigates to the monitoring dashboard.
Pre-condition	The admin must be logged into their account.
Postcondition	The admin has successfully monitored the platform and identified any issues.
Normal flow	The admin accesses the monitoring dashboard. The system displays real-time data on platform performance. The admin reviews the data and identifies any issues.
Alternative flow	None
Exceptions	Network connection issues
Business Rules	Monitoring data must be updated in real-time.

The Table 3.10 **Manage User Accounts** use case describes how an admin controls and maintains user accounts within the Skill Forge application. This process begins when an admin, who is already logged into their account, navigates to the user management section. The system recognizes this action as a trigger and provides access to all registered user accounts.

Once inside the user management interface, the system displays a complete list of users, including students and instructors. The admin can then perform various actions such as creating new accounts, modifying existing user details, or deleting accounts when necessary. These actions allow the admin to ensure that all user information remains accurate and up to date.

After the admin performs any operation, the system processes the request and updates the user accounts accordingly. Although there is no alternative flow, network-related issues may interrupt the process. Additionally, the system maintains a record of all changes made to user accounts for audit purposes, ensuring transparency, security, and accountability in managing user data.

The system also ensures that role-based access control is maintained while managing accounts, preventing unauthorized privilege assignments. Input validations are applied during updates to maintain data consistency and avoid errors. The admin can quickly search or filter users to efficiently manage large

datasets. This structured approach improves administrative efficiency and ensures reliable user account management across the platform.

Table 3.10 Manage User Account

Use case id	10
Use case name	Manage user accounts
Actors	Admin
Description	An admin manages user accounts, including creation, modification, and deletion.
Trigger	Admin intends to navigates to the user management section.
Pre-condition	The admin must be logged into their account.
Postcondition	User accounts are managed successfully.
Normal flow	The admin selects the user management option. The system displays the list of user accounts. The admin performs the necessary actions (create, modify, delete). The system updates the user accounts accordingly.
Alternative flow	None
Exceptions	Network connection issues.
Business Rules	User account changes must be logged for audit purposes.

The Table 3.11 **Manage Courses** use case describes how an admin controls and maintains all courses available on the Skill Forge platform. This process begins when an admin, who is already logged into their account, navigates to the course management section. The system recognizes this action as a trigger and provides access to the complete list of courses offered on the platform.

Once the course management interface is opened, the system displays all existing courses along with their relevant details. The admin can then perform various operations such as creating new courses, modifying existing course content or details, and deleting courses that are no longer required. These actions help ensure that the course catalog remains updated, relevant, and aligned with academic or organizational needs.

After any changes are made, the system processes the request and updates the course information accordingly. Although there is no alternative flow, network connectivity issues may temporarily interrupt the process. The system also logs all course-related changes for audit purposes, ensuring transparency, accountability, and proper tracking of administrative actions within the platform.

The system also validates course data before saving any updates to ensure consistency and completeness of information. The admin can manage course structure, including modules and associated resources, to maintain organized content delivery. Search and filtering options are available to quickly locate specific courses within a large catalog. This structured management approach improves efficiency and ensures the platform maintains high-quality and well-organized learning content.

Table 3.11 Manage Courses

Use case id	11
Use case name	Manage Courses
Actors	Admin
Description	An admin manages the courses offered on the platform, including creation, modification, and deletion.
Trigger	Admin intends to navigates to the course management section.
Pre-condition	The admin must be logged into their account.
Postcondition	Courses are managed successfully.
Normal flow	The admin selects the course management option. The system displays the list of courses. The admin performs the necessary actions (create, modify, delete). The system updates the courses accordingly.
Alternative flow	None
Exceptions	Network connection issues.
Business Rules	Course changes must be logged for audit purposes.

The Table 3.12 **Browse Course** use case describes how a student explores the available courses on the Skill Forge platform. This process begins when a student, who is already logged into their account, navigates to the course browsing section. The system recognizes this action as a trigger and loads the list of all available courses.

Once the section is opened, the system displays a structured list of courses along with relevant details such as course title, description, and category. The student can easily go through the available options and use filtering and sorting features to refine the search according to their interests, skill level, or subject area. This helps the student quickly find suitable courses that match their learning goals.

If the student is not logged in, the system redirects them to the login page to ensure proper authentication before accessing course information. This use case provides a simple and efficient way for students to discover and explore learning opportunities within the platform.

Additionally, the system ensures that course data is updated in real time so that students always view the latest available content. The student can select a course to view more detailed information before making a decision. The browsing interface is designed to be user-friendly, allowing smooth navigation across different categories. This structured approach enhances usability and helps students efficiently explore and select appropriate courses.[\[7\]](#)

Table 3.12 Browse Courses

Use case id	12
Use case name	Browse course
Actors	Student
Description	A student browses through the available courses.
Trigger	Student intends to navigates to the course browsing section.
Pre-condition	The student must be logged into their account.
Postcondition	The student has successfully viewed the list of available courses.
Normal flow	The student navigates to the "Browse Course" section. The system displays a list of available courses. The student can filter and sort the courses as needed.
Alternative flow	If the student is not logged in, the system redirects them to the login page.

3.3 Functional Requirements

The functional requirements of the Skill Forge project are designed to facilitate a dynamic, engaging, and efficient e-learning environment for students and professionals. These features not only aim to enhance student engagement but also ensure seamless learning, communication, and course management.

3.3.1 Course Enrollment

The course enrollment feature allows users to browse through a catalog of available courses, including free and paid options. Each course will include detailed information, such as course objectives, syllabus, pricing, and instructor details. Users will be able to sign up for courses effortlessly, with a step-by-step enrollment process that includes prompts and instructions to ensure smooth user experience.

Bulk enrollment will also be supported for users who wish to enroll in multiple courses at once. Additionally, the platform will provide course recommendations based on user interests and past enrollments, enhancing the user's learning path.

Users will also be able to view their enrollment status directly from their account dashboard for better tracking. The enrollment interface will maintain a simple and organized layout to reduce confusion during the registration process. Confirmation messages will be displayed after successful enrollment to clearly inform users that the selected course has been added to their learning profile.

Table 3.13 Course Enrollment

Identifier	FR-1
Title	Course Enrollment
Requirement ID	FR-1
Requirement	The system shall allow users to browse and enroll in available courses including free and paid options with complete course details.
Source	User Requirement
Rationale	To enable smooth onboarding and access to learning content.
Business Rule	Only authenticated users can enroll in courses.
Dependencies	Database, Authentication Module, Course Module
Priority	High

3.3.2 Progress Tracking

The platform will track users' progress through courses, displaying completed lessons, quizzes, and assignments. Users will have a dashboard that shows their learning journey and achievements.

The progress tracking feature provides users with a clear and structured view of their learning activities within the platform. It continuously updates the status of completed and pending lessons to help users stay organized. Visual indicators such as progress bars and completion percentages make it easy to monitor advancement at a glance. The system also records quiz scores and assignment submissions, allowing users to evaluate their performance over time. Notifications and reminders can guide users to continue incomplete courses and maintain consistency in learning. This feature supports goal setting by enabling users to track milestones and achievements. It also helps in identifying areas where additional practice or

revision is needed. Instructors and administrators can use this data to monitor user engagement and provide necessary support.

Table 3.14 Track Progress

Identifier	FR-2
Title	Progress Tracking
Requirement ID	FR-2
Requirement	The system shall track user progress across lessons, quizzes, and assignments and display it in a dashboard.
Source	User Requirement
Rationale	To help learners monitor their learning journey and achievements.
Business Rule	Progress data must update in real-time after each activity.
Dependencies	User Database, Course Module, Analytics Module
Priority	High

3.3.3 Real-Time Communication

The in-app chat module will enable real-time communication between students and instructors, providing immediate feedback and support. This feature will include text, voice, and possibly video chat options.

Table 3.15 Real-Time Communication

Identifier	FR-3
Title	Real-Time Communication
Requirement ID	FR-3
Requirement	The system shall allow real-time communication between students and instructors using chat features.
Source	User Requirement
Rationale	To provide instant doubt solving and interaction.
Business Rule	Only enrolled users can initiate communication.
Dependencies	Messaging Service, Authentication Module
Priority	High

3.3.4 Quizzes and Assessments

Quizzes and assessments will be embedded throughout the course content to test learners' knowledge and understanding of the material. Automated grading will be provided for multiple-choice or true/false quizzes, with instant feedback on correct and incorrect answers. Timed quizzes and assignments will also be available to assess the learners' ability to apply knowledge within specific time constraints.

Detailed reports will be provided after each assessment, showing a breakdown of areas of strength and improvement. Additionally, peer assessments and project-based evaluations will be incorporated to enhance practical learning.

Table 3.16 Quizzes and Assessments

Identifier	FR-4
Title	Quizzes and Assessments
Requirement ID	FR-4
Requirement	The system shall provide quizzes with automated grading, instant feedback, and result evaluation.
Source	User Requirement
Rationale	To evaluate learner understanding of course content.
Business Rule	Users must complete required lessons before attempting quiz.
Dependencies	Quiz Engine, Database, Course Module
Priority	High

3.3.5 Feedback and Reviews

At the end of each course, users will have the option to leave feedback and reviews. This will allow instructors to improve their course materials and adjust teaching methods. Reviews will include a rating system (e.g., 1 to 5 stars), along with space for detailed feedback.

Instructors will be able to respond to reviews, either addressing any concerns or thanking students for their suggestions.

This feature encourages active user participation by giving learners a voice in evaluating course quality. It helps maintain transparency by allowing new users to view honest opinions before enrolling in a course. The feedback collected can be analyzed to identify common issues and areas of improvement. Overall, it contributes to enhancing course effectiveness and user satisfaction on the platform.

Table 3.17 Feedback and Reviews

Identifier	FR-5
Title	Feedback and Reviews
Requirement ID	FR-5
Requirement	The system shall allow users to submit ratings and reviews after course completion.
Source	User Requirement
Rationale	To improve course quality and instructor performance.
Business Rule	Only completed course users can submit feedback.
Dependencies	Review Module, Course Module, Database
Priority	Medium

3.3.6 Mobile-Friendly Interface

The platform will be fully optimized for mobile devices, ensuring smooth and touch-friendly navigation on Android devices. The mobile version will be designed for seamless usage, including fast loading times, offline functionality (so students can access course materials without an internet connection), and responsive layouts that adjust to various screen sizes.

Table 3.18 Mobile-Friendly Interface

Identifier	FR-6
Title	Mobile-Friendly Interface
Requirement ID	FR-6
Requirement	The system shall provide a responsive and mobile-optimized interface for seamless learning on devices.
Source	System Requirement
Rationale	To ensure accessibility across mobile platforms.
Business Rule	UI must adapt to different screen sizes.
Dependencies	Frontend UI Framework
Priority	High

3.3.7 Administrative Tools

Administrators will have access to a set of administrative tools through the Android app to efficiently manage the platform. These tools will include options for user management (e.g., adding, removing, or updating student profiles), content upload and editing (e.g., uploading new courses, videos, and

resources), and course analytics (e.g., monitoring user engagement, completion rates, quiz scores).

Table 3.19 Administrative Tools

Identifier	FR-7
Title	Administrative Tools
Requirement ID	FR-7
Requirement	The system shall provide admin features for managing users, courses, and analytics.
Source	Admin Requirement
Rationale	To enable platform control and monitoring.
Business Rule	Only admin roles can access these tools.
Dependencies	Admin Module, Database, Analytics Engine
Priority	High

3.3.8 SMeet Live Session

The SMeet Live Session feature will enable learners to interact directly with instructors through scheduled real-time sessions, providing a more engaging and personalized learning experience. Users will be able to browse available instructors.

Table 3.20 SMeet Live Session

Identifier	FR-8
Title	SMeet Live Session
Requirement ID	FR-8a
Requirement	The system shall allow users to book and attend live sessions with instructors.
Source	User Requirement
Rationale	To enable real-time guided learning support.
Business Rule	Session booking requires availability validation.
Dependencies	Scheduling System, Notification Module, Video/Meeting
Priority	High

3.3.9 RAG-Based Assistant

The RAG-Based Assistant will provide an intelligent query-handling system that allows users to ask questions related to course videos and receive accurate, context-aware responses. This feature will utilize

video transcripts and metadata as its primary knowledge source, ensuring that responses are directly linked to the learning content. When a user submits a query, the system will process it and retrieve the most relevant sections of the video transcript using semantic search techniques.

The assistant will then generate a response based on the retrieved context, ensuring that answers remain precise and grounded in the actual course material. This approach minimizes irrelevant or incorrect information and improves the reliability of responses. Users will be able to interact with the assistant multiple times, enabling continuous clarification of concepts without needing instructor intervention. This feature supports self-paced learning, enhances accessibility, and provides instant academic assistance.

Table 3.21 RAG-Based Assistant

Identifier	FR-9
Title	RAG-Based Assistant
Requirement ID	FR-8b
Requirement	The system shall provide an AI assistant that answers queries using video transcripts and retrieval-based context.
Source	User Requirement
Rationale	To support intelligent learning assistance.
Business Rule	Responses must be generated only from relevant course content.
Dependencies	AI Model, Vector Database, Transcript Processor
Priority	High

3.3.10 Personalized Roadmap Generation

The Personalized Roadmap Generation feature will create structured learning paths tailored to each user's existing skills and learning goals. Users will input the skills they already possess, and the system will analyze this information to determine their current proficiency level. Based on this analysis, the platform will generate a roadmap divided into Beginner, Intermediate, and Advanced stages, ensuring a logical progression of learning.

Each stage of the roadmap will include recommended topics and learning sequences designed to help users gradually build expertise. The generated roadmap will be stored within the system, allowing users to revisit and track their progress over time. This feature will also adapt to different skill sets, making it flexible and scalable for a wide range of learners. By providing a clear and personalized direction, the

roadmap enhances learning efficiency and helps users achieve their educational goals in a structured manner.

Table 3.22 Personalized Roadmap Generation

Identifier	FR-10
Title	Personalized Roadmap Generation
Requirement ID	FR-9
Requirement	The system shall generate a personalized learning roadmap based on user skills.
Source	User Requirement
Rationale	To guide learners through structured skill progression.
Business Rule	User must input valid skill data before generation.
Dependencies	Recommendation Engine, User Profile Module, Database
Priority	High

3.4 Non-Functional Requirements

Nonfunctional requirements are those that establish parameters that may be used to evaluate the performance of a system. Nonfunctional requirements are the limitations to which the system will work.

The non-functional requirements of our project encompass various aspects related to usability, performance, security, portability, reliability, and other quality attributes. Here are some key non-functional requirements.[\[1\]](#)

3.4.1 Usability

Skill Forge is designed with a focus on usability to ensure that users of all technical backgrounds can effectively use the platform. Key usability features include:

- **Intuitive Design:** The user interface (UI) will be intuitive, with clear icons, labels, and instructions to guide users through the platform. Each screen and layout will follow consistent design principles and color schemes, making navigation predictable and comfortable even for first-time users. Buttons and input fields will be prominently placed to prevent user confusion and minimize learning curves.
- **Responsive Design:** The platform will be responsive, adapting seamlessly to different screen sizes and orientations, providing an optimal experience on both smartphones and tablets. Whether users access the app in portrait or landscape mode, all content and features will adjust dynamically to maintain readability, accessibility, and full functionality without horizontal

scrolling.

- **User-Friendly Navigation:** Simplified and consistent navigation menus will help users easily find and access courses, forums, and other features. Menus will be logically grouped, with expandable sections and icons for quick recognition. Frequently accessed pages like “My Courses,” “Dashboard,” and “Messages” will be readily available via bottom navigation or side drawers.
- **Search Functionality:** An advanced search feature will allow users to quickly find courses, topics, and resources. Search will support filtering by category, course level, instructor, and rating, and include smart suggestions based on user behavior and trending topics. Results will be displayed in a clean, organized manner with thumbnails and brief descriptions for quick scanning.
- **Feedback Mechanisms:** Users will have the ability to provide feedback on their experience, helping to continuously improve the platform’s usability. Feedback options will include in-app surveys, rating systems, and suggestion boxes. All submissions will be reviewed regularly by the development team to address issues and implement enhancements.
- **Performance Optimization:** The platform will be optimized for performance, ensuring quick load times and smooth interactions, even under varying network conditions. Images and videos will be compressed and cached efficiently, and asynchronous data loading will prevent lags during user interaction. Background tasks will be managed to avoid unnecessary battery drain on mobile devices.
- **Error Handling and Help:** Clear error messages and help options will be provided to assist users in resolving any issues they encounter. Context-sensitive help will offer guidance based on the user’s current activity. For example, if a course fails to load, the user will be shown the cause and steps to retry or contact support. A dedicated support chat or ticketing system will also be available for more complex issues.

3.4.2 Portability

- Skill Forge should be compatible with a wide range of devices and operating systems, including smartphones, tablets, and for different screen sizes. It will support Android versions from 8.0 and above, ensuring accessibility to users with varying device capabilities.
- The app should adapt gracefully to different screen sizes, resolutions, and orientations, ensuring a consistent user experience across devices. A responsive design framework, along with scalable vector graphics and flexible layouts, will ensure visual and functional consistency across all supported devices.

3.4.3 Security

- Data privacy and protection should be ensured through robust security measures, including encryption of sensitive information and secure authentication mechanisms. All user data, including passwords and payment details, will be encrypted using industry-standard protocols.
- User authentication and authorization processes should be implemented securely to prevent unauthorized access to user accounts and data. Two-factor authentication (2FA), session management, and token-based authorization will be integrated to enhance user account protection.
- The app should comply with relevant data protection regulations and standards to safeguard user privacy. This includes adherence to GDPR and other regional compliance frameworks, maintaining detailed audit trails, and providing users with control over their personal data.

3.4.4 Performance

- The app should be responsive and provide quick loading times, even during peak usage periods. Optimized backend processes, efficient database queries, and content delivery networks (CDNs) will be utilized to minimize latency and improve performance.
- Response times for actions such as course registration, progress checking, and content loading should be minimal. Skill Forge will use caching strategies, preloading techniques, and asynchronous processing to ensure real-time responsiveness and seamless user interaction.
- The system should be able to handle a large number of concurrent users without significant degradation in performance. Load balancing techniques and stress testing will be employed to ensure stable performance under high traffic conditions. Monitoring tools will be in place to detect and resolve performance bottlenecks quickly.

3.4.5 Reliability

- Skill Forge should be compatible with a wide range of devices and operating systems, including smartphones, tablets, and for different screen sizes. It will support Android versions from 8.0 and above, ensuring accessibility to users with varying device capabilities.
- The app should adapt gracefully to different screen sizes, resolutions, and orientations, ensuring a consistent user experience across devices. A responsive design framework, along with scalable vector graphics and flexible layouts, will ensure visual and functional consistency across all supported devices.

3.4.6 Scalability

- The Skill Forge platform is designed to scale effectively to accommodate a growing user base and increasing data volumes over time. Modular architecture and microservices will be employed to isolate functions and make them independently scalable.
- The architecture supports both horizontal and vertical scaling to handle spikes in user activity and resource demands. Cloud-based solutions such as Kubernetes and auto-scaling groups will be utilized to dynamically allocate resources as needed.
- This scalable design ensures a smooth and uninterrupted learning experience as more users and content are added. System performance will remain stable with consistent load balancing and optimized resource management.
- The platform can expand seamlessly while maintaining high performance and reliability. Periodic performance assessments and continuous integration/continuous deployment (CI/CD) pipelines will support ongoing enhancements without impacting live operations.

Chapter 4

Design and Architecture

4. Design and Architecture

4.1 System Architecture

The Skill Forge e-learning platform is built on a modular microservices architecture designed for scalability, security, and a seamless user experience. The backend, developed in Kotlin, consists of independent microservices that handle core operations such as course enrollment, progress tracking, quizzes, and real-time features like chat and notifications. Structured data is stored in a relational database, while NoSQL and cloud storage solutions manage unstructured content and large media files. Cloud hosting services such as AWS or Google Cloud ensure high availability and horizontal scalability. Real-time communication is powered by Firebase, and security measures including data encryption and rate limiting protect user information. An admin dashboard provides centralized management, while integrated analytics tools track user engagement and overall platform performance. This architecture delivers a flexible, reliable, and secure e-learning experience for all users.

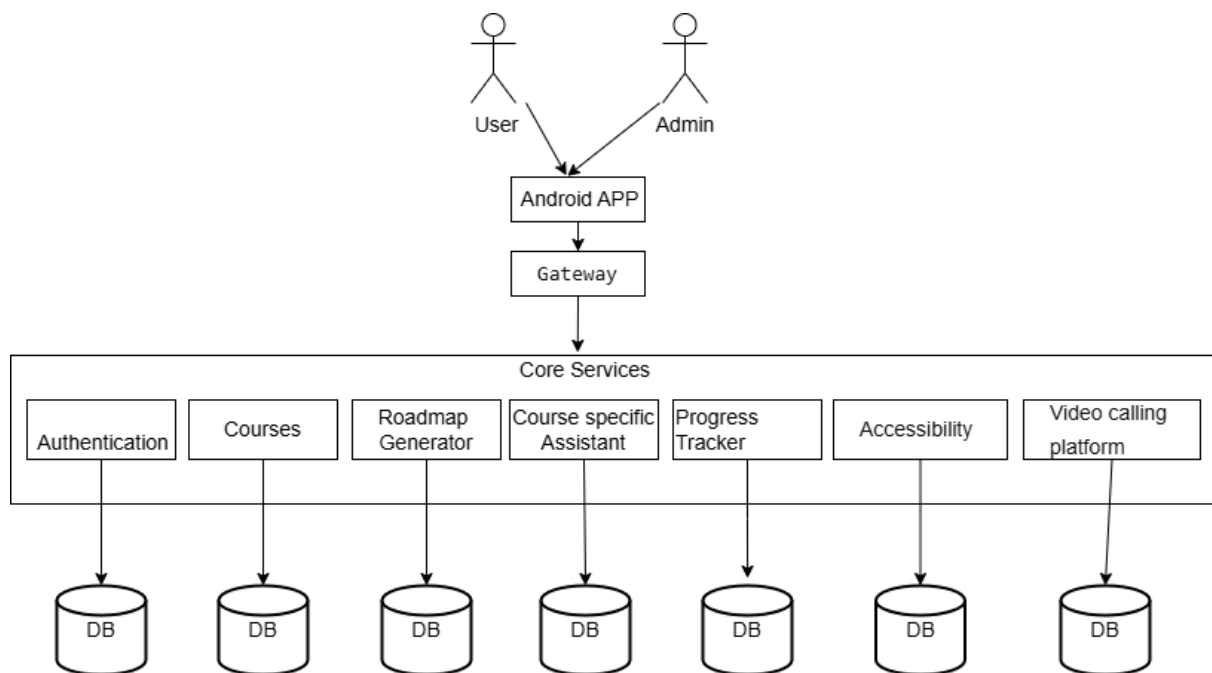


Figure 4.1 Architecture Diagram

The architecture of Skill Forge is designed to ensure scalability, performance, and a rich user experience across both mobile and web platforms. At the core of the system lies the backend, which handles all the business logic and data operations.

Cloud storage services are used to securely store media files such as course videos, and documents, ensuring they are readily accessible and scalable.

The backend is hosted on cloud platforms like firebase, which provide the necessary infrastructure for performance and scalability. A layer acts as the bridge between the frontend and backend, managing all

user requests including course enrollment, quiz submissions.

This layer communicates with the database, which may use a combination of relational databases for structured data like user accounts and course records, and NoSQL solutions for dynamic data like user activity logs or chat messages.

To support real-time interactions such as live chat between students and instructors or instant notifications about quizzes, Firebase is integrated into the system. The frontend of Skill Forge, built with mobile compatibility (e.g., Kotlin for Android) and potentially a responsive web interface, connects to display user-specific content such as available courses, quiz results. It also handles real-time updates and push notifications to keep users informed.

This panel allows admins to manage users, upload or modify courses, view quiz statistics. Additionally, it provides access to analytics tools, which track user engagement, platform performance, and content effectiveness, feeding insights back to the admin dashboard to help guide improvements.

Overall, the system architecture of Skill Forge is built to support a seamless, interactive, and secure e-learning experience, complete with real-time capabilities and robust administrative tools.

4.2 Process Flow/Representation

The Fig 4.2 user flow diagram illustrates the complete journey of a learner within the **Skill Forge** application from login to course completion. The process begins when the user opens the application and enters login credentials. The system then checks whether the provided credentials are valid. If the credentials are incorrect, the user is shown an error message and must try again. If the credentials are valid, the user is directed to the dashboard, where learning activities begin. From the dashboard, the user can enroll in a course and start watching video lessons that deliver the learning content in a structured sequence.

After viewing the lessons, the system checks whether the user requires assistance. If help is needed, the learner can interact with the **AI Smart Assistant**, which provides guidance during the learning process. If no help is required, the user continues learning directly. The flow then asks whether the learner wants to join a live session. If the user chooses to participate, they proceed to the session for real-time interaction, while users who decline continue to the next stage without interruption. This flexible path allows the application to support both independent and interactive learning preferences.

Once the learning activities are completed, the user can access the progress dashboard to review performance and then take a quiz to evaluate understanding of the course material. After the quiz, the application tracks the learner's progress and records current skill levels. Based on this information, the system generates a personalized roadmap that recommends future learning paths according to the user's

development. Finally, the learner can provide feedback about the experience before reaching the end of the process. This user flow demonstrates a structured learning journey that combines authentication, guided learning, support, assessment, progress monitoring, and personalized improvement within a single application.

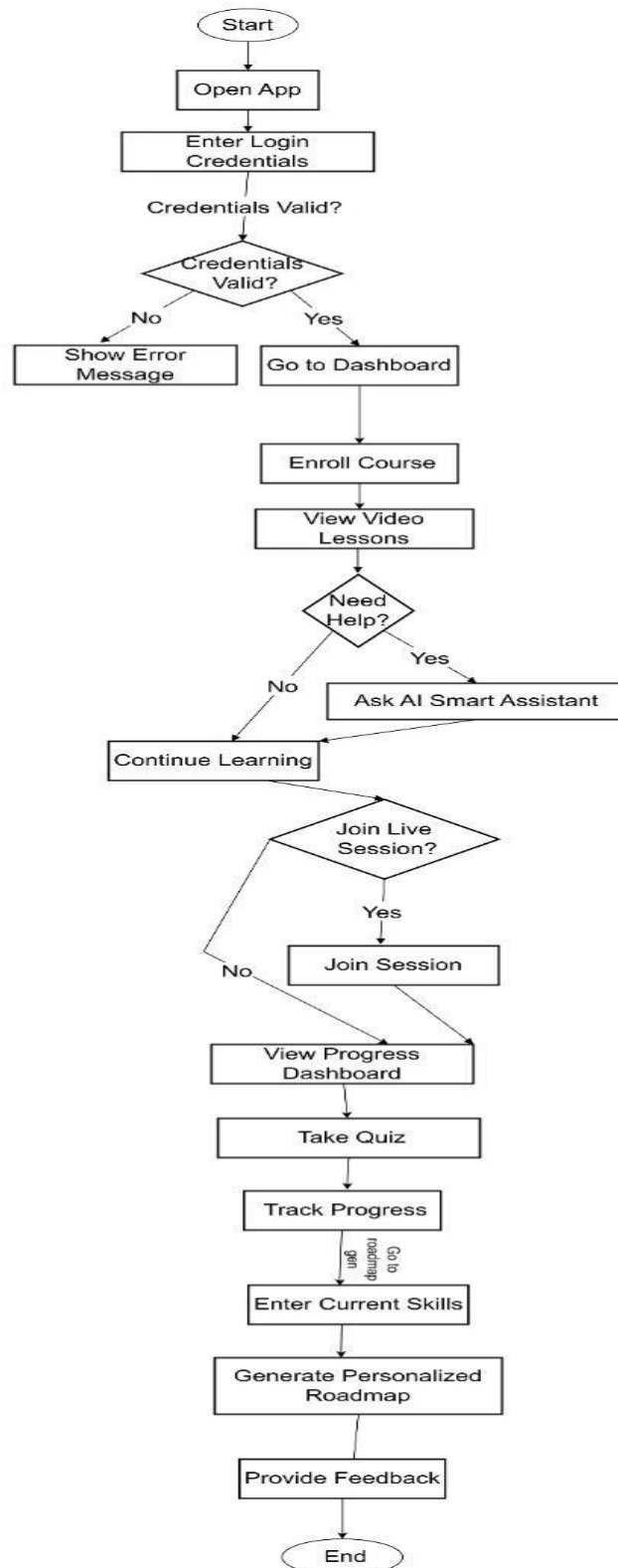


Figure 4.2 User Flow Diagram

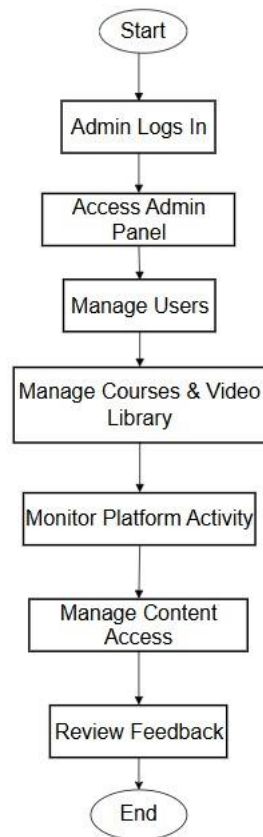


Figure 4.3 Admin Flow

The Fig 4.3 admin flow diagram illustrates the sequence of actions performed by the administrator within the **Skill Forge** application to manage the platform effectively. The process begins when the administrator logs into the system and gains access to the admin panel, which serves as the central control area for platform operations. From this panel, the administrator can manage user accounts by monitoring user information and controlling account-related activities. The next stage allows the administrator to manage courses and the video library, ensuring that learning materials remain organized and updated for users.

After managing course content, the administrator monitors overall platform activity to observe system performance and user engagement across the application. The flow then continues to content access management, where the administrator can control permissions and ensure that users can access appropriate learning resources. Following this, the administrator reviews feedback submitted by users to understand their experience and identify areas that may require improvement. Once these administrative tasks are completed, the process reaches its endpoint. This flow demonstrates a structured administrative process that supports user management, content control, system monitoring, and continuous platform improvement within the application.

4.2.1 Process Flow Description

The Skill Forge app begins its user journey the moment a user opens the app. The first interaction requires users to enter their login credentials, which are then validated by the system against stored user data. This step ensures secure access control.

If the credentials are valid, the user is seamlessly directed to the dashboard, their central hub for learning and career development activities. If invalid, the system displays an error message, guiding the user to re-enter correct credentials or recover their account.

Once inside the dashboard, users can access the course catalog, which displays a categorized list of available courses based on topics, levels, or career paths. Users can browse through these courses, reading descriptions, learning outcomes, and reviews.

After finding a suitable course, the user selects the course and initiates enrollment, which might involve payment (if it's a paid course) or simply confirmation (for free ones). After successful enrollment, the course content is made available, including video lectures, reading materials, assignments, and other interactive resources.

With content access granted, users can also navigate to the quiz or assessment section, which is embedded within the course structure. They can take quizzes to evaluate their understanding of the subject.

After completing a quiz, they submit it, and the backend automatically evaluates the quiz using predefined correct answers or criteria. The results are then shown, often with explanations or feedback for each question to enhance learning.

Additionally, Skill Forge supports real-time chat, enabling instant communication with peers, instructors, or support staff particularly useful for doubt clearance or community interaction. After completing a course, learners are encouraged to provide feedback, which is valuable for both platform improvement and helping other users choose quality content. Upon completing all course components, the system recognizes that the user has successfully completed the course and gained knowledge, possibly awarding certificates or badges.

Parallel to the learner journey, the admin flow plays a vital role in managing the platform. Admins log in to a dedicated portal, the Admin Panel, built using Django. From here, they can manage users, including on-boarding, roles, permissions, and behavior monitoring.

They can also manage courses, including adding new content, updating materials, and organizing course structure. Course enrollments are tracked and managed, allowing admins to view student

progress, handle enrollment limits, or resolve issues.

A crucial feature of Skill Forge is the integration of analytics tools, which allow admins to view insights such as user activity, quiz performance, content popularity, and platform engagement. These insights help in decision-making and platform optimization.[\[4\]](#)

4.3 Sequence Diagram

A sequence diagram is a type of UML (Unified Modeling Language) interaction diagram that shows how objects interact with each other in a time-sequenced order. It visually represents the flow of messages between various system components or objects to accomplish a specific functionality or use case.

Each object in the diagram is represented by a lifeline, and messages exchanged between these lifelines are depicted as arrows, often annotated with the message name and parameters.

Sequence diagrams are particularly useful for understanding dynamic behavior, such as the order of operations, method calls, responses, and object interactions. They are frequently used during the design phase of software development to model real-time scenarios, track how a particular task is performed, and ensure that all necessary steps and interactions are considered.

The diagram also helps in identifying potential design flaws early, such as tight coupling, unnecessary dependencies, or violations of object responsibilities.

Common elements include activation bars (indicating when an object is active), loops, conditions, parallel flows, and object creation or deletion. They are widely used in modeling client-server interactions, web service communications, event-driven systems, and more .

Sequence diagrams also support clear communication among developers, designers, and stakeholders by providing a straightforward visual representation of system interactions. They make it easier to trace the exact sequence of operations required to complete a process. By illustrating message flow, they help ensure that system logic aligns with functional requirements. These diagrams can be refined iteratively as the system design evolves. They are useful for validating use cases by mapping each step to actual system behavior. Additionally, they assist in debugging by identifying missing or incorrect interactions. Sequence diagrams can also serve as a reference during implementation and testing phases. They help maintain consistency between design and development. Overall, they contribute to building well-structured and understandable system workflows.

They also improve documentation quality by clearly outlining interaction details for future reference. This makes it easier for new team members to understand system behavior. As a result, they support long-term maintenance and scalability of the system.

4.4 Phase 1: User Side

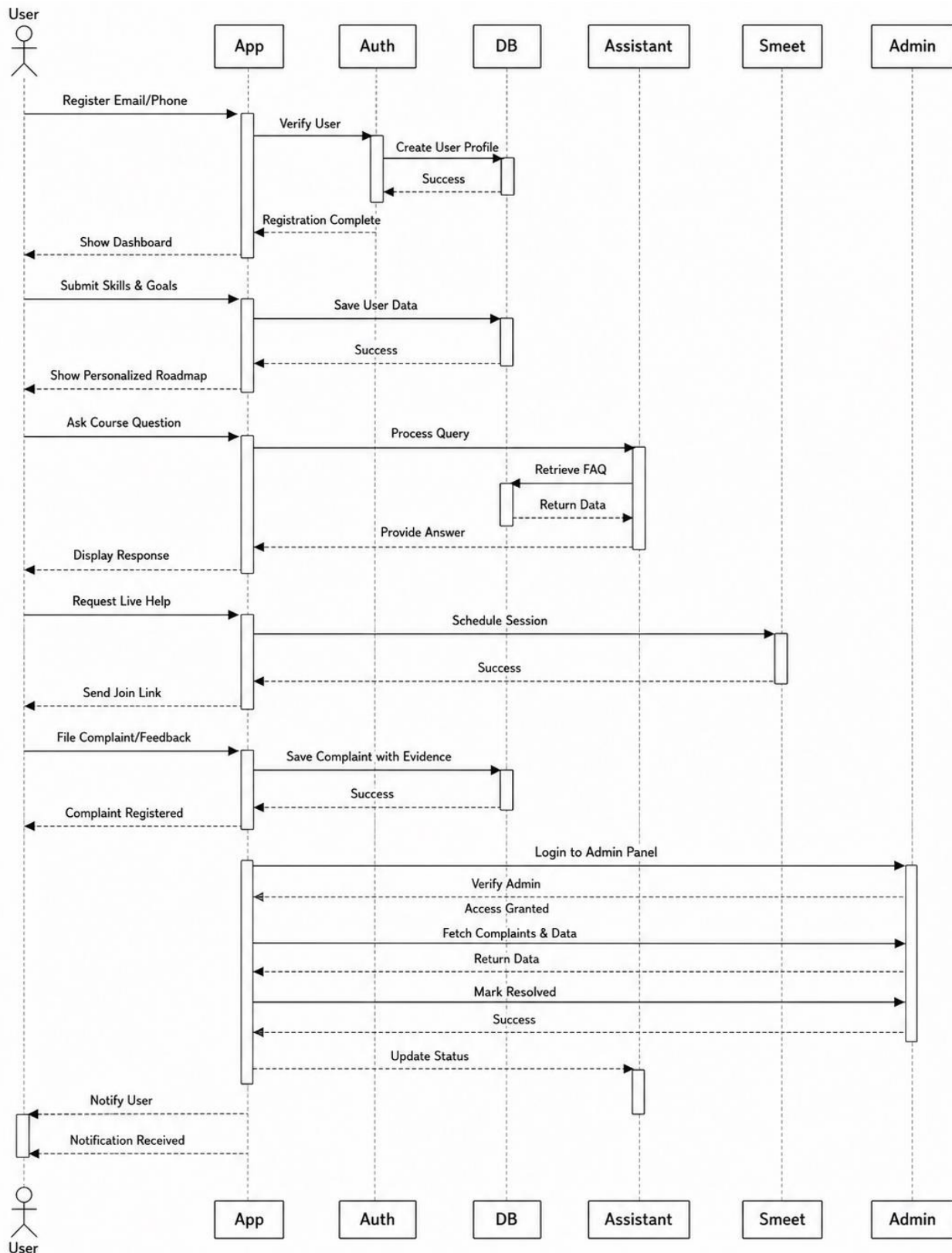


Figure 4.4 User Side Sequence Diagram

4.4.1 Sequence Diagram Description

The Skill Forge App serves as the central interface for the user. It acts as a coordinator, sending user-initiated actions to the relevant backend modules and receiving responses to update the UI.

The user begins by signing up or logging in, views courses, enrolls in them, watches lectures, attempts quizzes, submits feedback, sends messages to admin, and finally logs out all through this primary interface.

It ensures a seamless and intuitive user experience by integrating all modules into a unified platform. The app is designed to be responsive and accessible across various devices, making learning and interaction convenient anytime, anywhere.

4.4.2 Authentication Module

The Authentication Module is the backbone of Skill Forge's user security, responsible for handling all registration and login operations across the platform. When a new user signs up, the module securely collects their information, validates input fields (such as email, phone number, and password), and creates a unique user profile in the database.

Once the account is created, a confirmation message is generated and users are directed to the next onboarding steps. For returning users, the login process involves validating credentials against securely stored encrypted data. If credentials are correct, a session is initiated, and the user gains access to their personalized dashboard and functionalities based on their role (student, instructor, or admin).

Beyond basic login and registration, this module also supports additional security features such as password reset via email, two-factor authentication (2FA), for mobile devices. It integrates seamlessly with other modules to control access permissions and ensure that users only interact with resources relevant to their role.

The Authentication Module is designed with scalability and compliance in mind, adhering to modern data protection regulations such as GDPR, to safeguard sensitive user information.

4.4.3 Courses Module

The Courses Module handles course-related functionalities. After login, when the user chooses to view available courses, this module retrieves and returns a list. Upon selecting a course, it processes the enrollment and confirms the user's registration. It is a crucial part of the learning journey as it connects users with educational content.

In addition to enrollment, the module manages course progression, allowing users to track their learning status, resume from where they left off, and view completion certificates upon finishing a

course. It supports filtering by category, difficulty level, and popularity to help users discover relevant content efficiently.

4.4.4 Video Player

The Video Player module streams lecture content seamlessly to enrolled users, ensuring smooth playback with automatic quality adjustment based on internet speed. It offers essential controls like play, pause, and volume adjustments, and supports multiple video formats for an optimal viewing experience.

Additionally, the module features subtitle support, video bookmarking, and resume options, enhancing user engagement. Integrated with progress tracking and assessments, it reflects user progress, ensuring a cohesive and uninterrupted learning experience.

4.4.5 Feedback System

The Feedback System in Skill Forge is designed to collect, process, and manage user reviews and opinions regarding courses, instructors, and the overall learning experience. After completing lectures or assessments, users are prompted to provide feedback through structured forms that include ratings, comments, and suggestions.

This input is stored in the system and linked to the respective course or instructor for easy access and analysis. The feedback mechanism is user-friendly and encourages honest reflection by ensuring anonymity when needed and offering incentives like badges or recognition for active reviewers.

This module plays a vital role in maintaining and improving content quality across the platform. Administrators and instructors can regularly review the submitted feedback to identify strengths and weaknesses in course delivery, structure, and content.

Constructive criticism helps instructors adapt their teaching methods, while positive feedback reinforces effective practices. Over time, the system builds a valuable repository of user insights, which supports data-driven decision-making and enhances overall user satisfaction by ensuring the learning environment evolves in response to learners' needs.

4.4.6 Quiz Engine

The Quiz Engine is a robust module that manages all quiz-related functionalities within the platform. It is responsible for delivering quizzes to users, capturing their responses, and evaluating the results in real time.

When a user attempts a quiz, this module efficiently processes the submitted answers and provides instant feedback, highlighting correct and incorrect responses. This immediate result generation helps learners assess their performance on the spot and identify areas for improvement.

In addition to basic quiz-taking capabilities, the Quiz Engine supports various question formats, such as multiple-choice, true/false, short answer, and matching types, catering to diverse learning styles. It also incorporates scoring logic, time tracking, and progress monitoring to ensure a comprehensive assessment experience.

4.4.7 Profile Manager

The Profile Manager is responsible for managing user profile data. Users can update their email or phone number, and this module processes those changes and returns a confirmation. It helps maintain accurate user information across the system.

Beyond basic updates, the Profile Manager ensures data integrity by validating input formats and preventing duplicate or invalid entries. It also allows users to view their activity history, manage security settings such as password changes, and upload profile pictures.

Integrated with other modules, it ensures that personalized information like enrolled courses or quiz history is accurately reflected in the user's dashboard, enhancing the overall user experience.

4.4.8 Chat System

The Chat System is an essential feature of Skill Forge, designed to foster real-time communication between users, instructors, and administrators. It supports both individual and group conversations, allowing learners to seek clarification, request guidance, or report technical issues directly within the app.

Furthermore, the Chat System includes administrative tools for managing conversations efficiently. Admins can prioritize messages, track user interactions, and maintain logs for transparency and quality control.

Future enhancements may include multimedia message support (voice notes, file attachments, images) and AI-based auto-reply for frequently asked questions. This module not only improves user experience by ensuring prompt support but also builds a stronger sense of community within the Skill Forge platform.

Additionally, users can directly contact support through the chat to request certificates after course completion. The system allows users to send queries related to certification status and receive timely updates. Users can also initiate help requests for any issues faced during learning or platform usage. A dedicated option enables quick messaging to the admin for urgent concerns or guidance. This functionality ensures that users remain supported and connected throughout their learning journey.

4.5 Phase 2: Admin Side Sequence Diagram

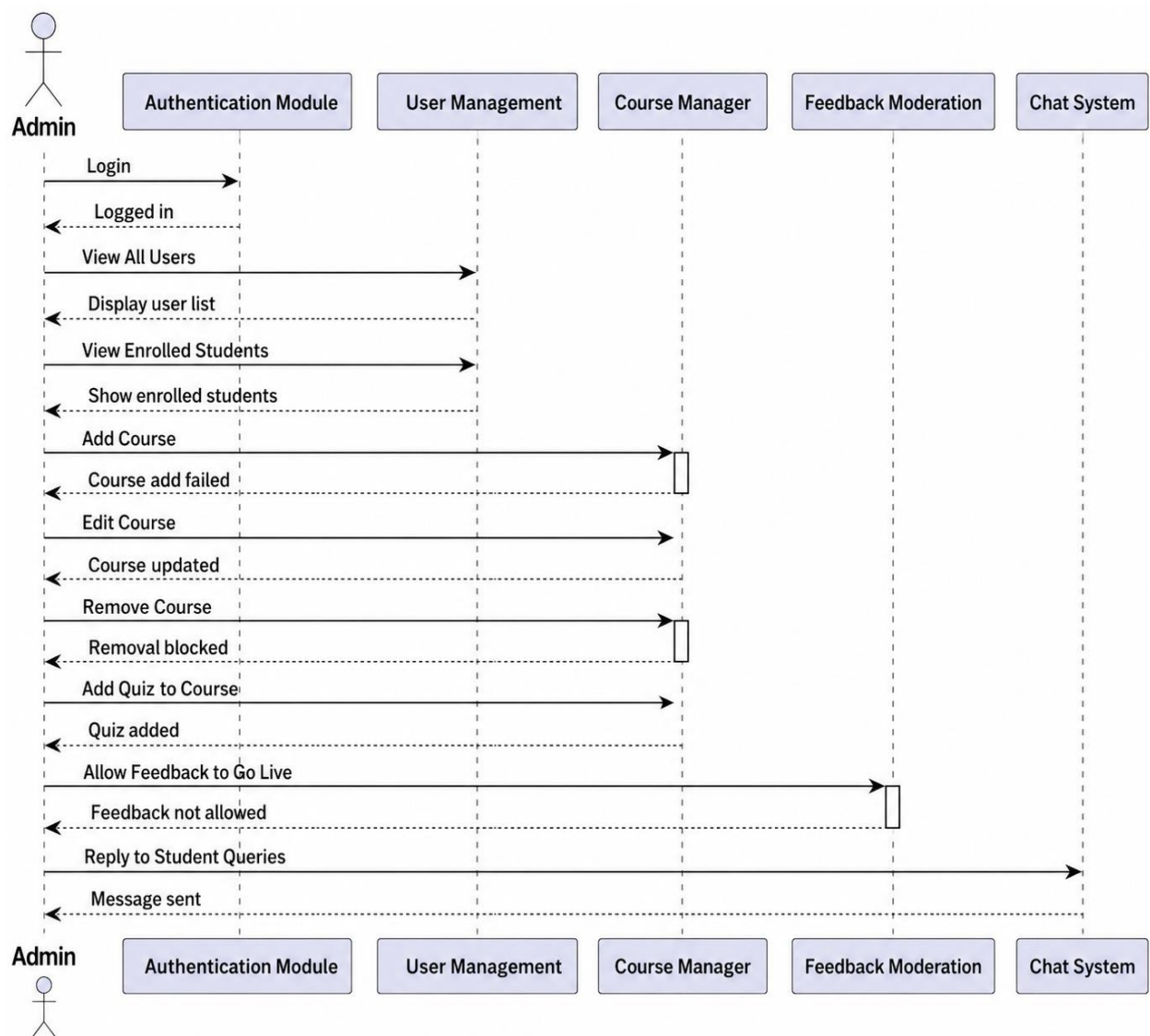


Figure 4.5 Admin Side Sequence Diagram

The Admin Panel is the central interface for administrators to manage the platform, including user accounts, courses, and quizzes. Admins can log in securely, add or remove users, update course content, create and grade quizzes, and manage feedback to ensure a smooth learning experience.

4.5.1 Authentication Module

The Authentication Module plays a crucial role in ensuring secure access to the admin panel. Upon the admin's login attempt, it verifies the provided credentials, such as the username and password, against the stored data. If the details match, the module grants access, ensuring that only authorized personnel can interact with the system's management features.

This module is designed with security in mind, employing encryption and secure authentication

mechanisms to protect sensitive admin data. By managing the authentication process, it prevents unauthorized access and maintains the integrity of the admin panel, ensuring that only legitimate users can perform administrative tasks.

4.5.2 User Management

The User Management module allows administrators to efficiently oversee all user accounts on the platform. Admins can access detailed user profiles, including account status, roles, and activity logs, enabling them to track user engagement. This functionality is crucial for monitoring the platform's user base and ensuring that all users follow the platform's terms of use.

Additionally, the module provides a focused view of enrolled students, allowing admins to track their course participation and progress. This oversight ensures that students are actively engaging with their learning materials and helps in identifying any issues that need attention. It plays a vital role in maintaining a smooth and organized learning environment.

4.5.3 Course Manager

The Course Manager module oversees the entire lifecycle of course content on the Skill Forge platform. It allows administrators and instructors to add new courses by inputting essential details such as title, description, objectives, pricing, and multimedia content. Courses can be organized by category and difficulty, making them easier for users to find.

Existing content can be updated or removed as needed, with confirmation prompts to avoid accidental changes. The system also ensures version control and maintains an organized structure for all active and archived courses.

Additionally, the module provides performance insights like enrollment numbers, course ratings, completion rates, and user feedback to help instructors refine content. Role-based access controls ensure that only authorized personnel can manage or modify courses, preserving content integrity. By offering centralized control and real-time updates, the Course Manager ensures a consistent, relevant, and engaging learning experience for all users on the platform.

4.5.4 Feedback Moderation

The Feedback Moderation module allows admins to review and manage student feedback before it becomes publicly visible. Admins have the ability to approve, edit, or reject feedback, ensuring that only constructive and relevant reviews are displayed on the platform. This process helps maintain the quality and integrity of the feedback system.

By controlling what feedback is shared publicly, this module fosters a positive environment where users feel confident sharing their experiences. It ensures that feedback contributes to the improvement of courses and overall platform quality while preventing inappropriate or harmful content from being

visible to others.

4.5.5 Chat System

The Chat System facilitates real-time communication between admins and students, allowing admins to respond promptly to queries and issues raised by users. This system ensures that students receive timely support, enhancing user satisfaction and engagement. Admins can view incoming messages and provide immediate responses, fostering an open channel for communication. Moreover, the system ensures that each admin's response is accurately delivered to the concerned student, maintaining a seamless flow of conversation. This two-way communication feature strengthens the support infrastructure of the platform, ensuring that students' concerns are addressed efficiently, thereby improving the overall user experience.

4.6 Class Diagram

This class diagram is a graphical representation of the system's classes, their attributes, and the relationships between them. It elaborates structure of the Skill Forge and shows how the different classes in the system are related to each other.

This class diagram represents the structure of the Skill Forge app, outlining its core classes, attributes, and relationships. At the center is the User class, which includes shared attributes like userID, userName, email, and methods for login, logout, and profile updates.

It branches into two main roles: Admin and Student. Admins can manage users, courses, enrollments, view analytics, and approve/upload internships. Students can enroll in courses, view transcripts, take quizzes, and apply for jobs.

Courses are managed by admins and include content and enrolled students. Each course may have quizzes, consisting of questions and evaluated results. The job and internship module is handled through the Job, System, and Community classes. Admins and the system upload jobs, notify users, and manage listings, while students and the community can view and apply for these opportunities.

Overall, the diagram shows how Skill Forge integrates learning and career services in a structured, role-based system.[\[6\]](#)

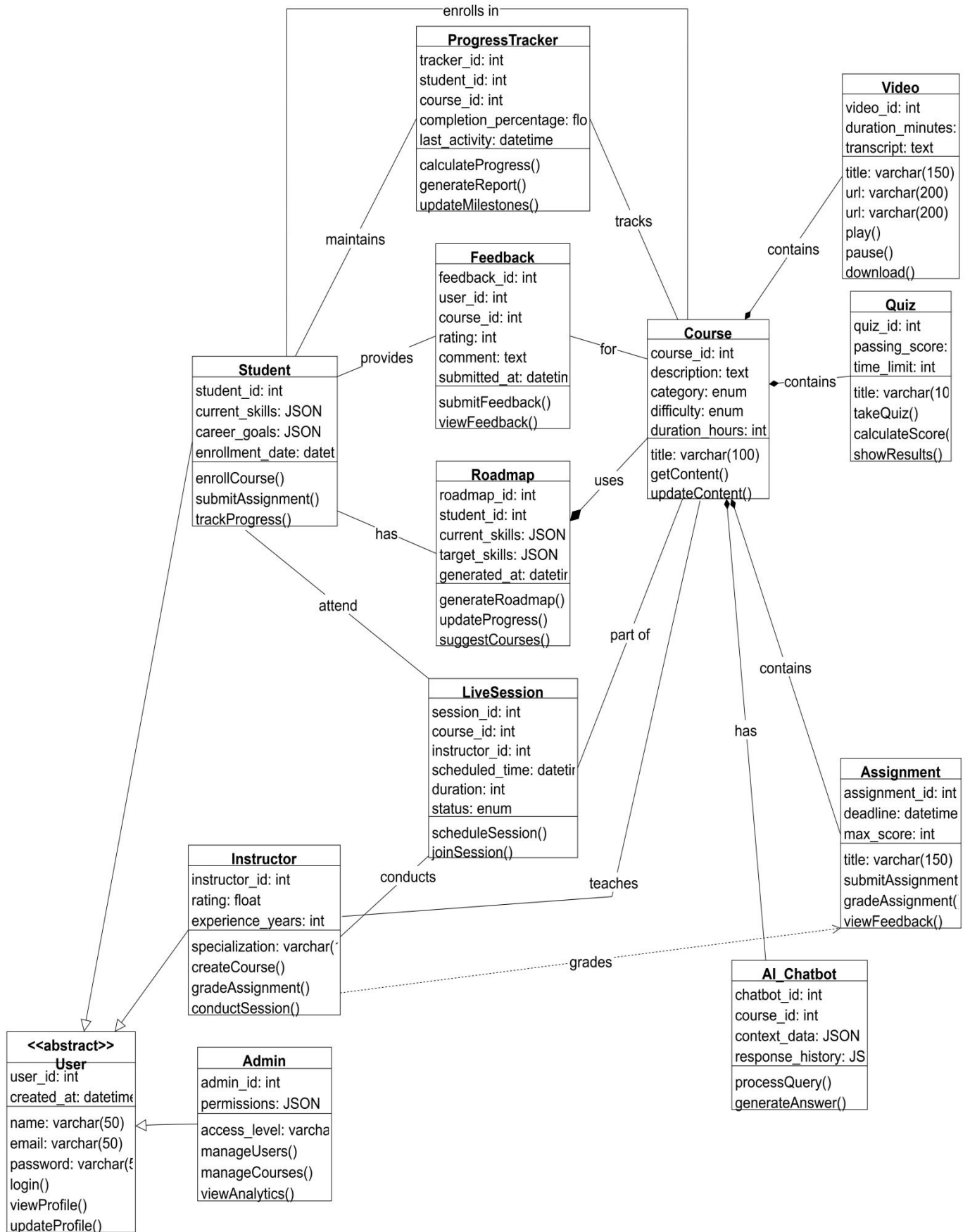


Figure 4.6 Class Diagram

4.6.1 Class Diagram Description

The class diagram illustrates the structure and functionality of an e-learning and course management system, showing how different components interact within the platform.

At the core is the abstract User class, which contains attributes such as `user_id`, `created_at`, `name`, `email`, and `password`. It provides basic operations including `login()`, `viewProfile()`, and `updateProfile()`. This class is extended by three specialized roles: Student, Instructor, and Admin.

The Student class represents learners in the system. It includes attributes such as `student_id`, `current_skills`, `career_goals`, and `enrollment_date`. Students can perform actions like `enrollCourse()`, `submitAssignment()`, and `trackProgress()`. A student maintains progress through the `ProgressTracker`, provides feedback using the `Feedback` class, has a `Roadmap`, and attends `LiveSession` instances.

The Instructor class extends User and includes attributes such as `instructor_id`, `rating`, `experience_years`, and `specialization`. It provides methods like `createCourse()`, `gradeAssignment()`, and `conductSession()`. Instructors teach courses, conduct live sessions, and grade assignments.

The Admin class also inherits from User and manages system-level operations. It includes attributes such as `admin_id`, `permissions`, and `access_level`, and provides functionalities like `manageUsers()`, `manageCourses()`, and `viewAnalytics()`.

The Course class is a central entity in the system. It includes attributes such as `course_id`, `description`, `category`, `difficulty`, `duration_hours`, and `title`. It provides methods like `getContent()` and `updateContent()`. A course contains Video, Quiz, and Assignment components, is associated with an `AI_Chatbot`, and is linked with `LiveSession`.

The Video class represents course content in video format, with attributes such as `video_id`, `duration_minutes`, `transcript`, `title`, and `URL`. It includes methods like `play()`, `pause()`, and `download()`.

The Quiz class handles assessments and includes attributes such as `quiz_id`, `passing_score`, `time_limit`, and `title`. It provides methods `takeQuiz()`, `calculateScore()`, and `showResults()`.

The Assignment class manages coursework submissions. It contains attributes such as `assignment_id`, `deadline`, `max_score`, and `title`, along with methods `submitAssignment()`, `gradeAssignment()`, and `viewFeedback()`.

The ProgressTracker class tracks a student's progress in a course. It includes attributes such as tracker_id, student_id, course_id, completion_percentage, and last_activity. Its methods include calculateProgress(), generateReport(), and updateMilestones().

The Feedback class stores user feedback for courses. It includes attributes such as feedback_id, user_id, course_id, rating, comment, and submitted_at, with methods submitFeedback() and viewFeedback().

The Roadmap class represents a learning path for students. It contains attributes such as roadmap_id, student_id, current_skills, target_skills, and generated_at. It provides methods generateRoadmap(), updateProgress(), and suggestCourses(), and is associated with Course.

The LiveSession class represents real-time sessions. It includes attributes such as session_id, course_id, instructor_id, scheduled_time, duration, and status. It provides methods scheduleSession() and joinSession(). Instructors conduct these sessions and students attend them.

The AI_Chatbot class is associated with courses and provides automated assistance. It includes attributes such as chatbot_id, course_id, context_data, and response_history, and provides methods processQuery() and generateAnswer().

Overall, the diagram shows how users interact with courses, assessments, progress tracking, live sessions, and feedback within the system.

Chapter 5

Implementation

5. Implementation

In the implementation phase of the Skill Forge e-learning app, we have developed algorithms that power the application's core features, ensuring a smooth and effective learning experience for students, and administrators.

5.1 Algorithm

5.1.1 User Authentication (Login/Signup)

Function: authenticateUser(email, password)

Input: email (String), password (String) **Output:**

user (Object) or error (String)

Pseudocode:

1. Retrieve user from database using the provided email.
2. If user exists:
 - a. Compare provided password with stored hashed password.
 - b. If matched, return the user object.
 - c. Else, return error: "Incorrect password".
3. If user doesn't exist, return error: "User not found".

Function: registerUser(email, password, username)

Input: email (String), password (String), username (String)

Output: success (Boolean) or error (String)

Pseudocode:

1. Check if email already exists.
2. If not:
 - a. Hash password and store user info in the database.
 - b. Return success.
3. Else, return error: "User already exists".

5.1.2 Enroll in Course

Function: enrollInCourse(user, courseId)

Input: user (Object), courseId (String)

Output: success (Boolean) or error (String)

Pseudocode:

1. Verify if the user is already enrolled.
2. If not:
 - a. Add entry in the user's enrolled courses list.
 - b. Return success.
3. Else, return error: "Already enrolled in this course".

5.1.3 Watch Lectures

Function: watchLecture(user, courseId, lectureId)

Input: user (Object), courseId (String), lectureId (String)

Output: videoURL (String) or error (String)

Pseudocode:

1. Check if user is enrolled in the course.
2. If yes:
 - a. Retrieve lecture URL from the database.
 - b. Return the video URL.
3. Else, return error: "Access denied. Please enroll first."

5.1.4 Take Quiz and Get Real-Time Results

Function: submitQuiz(user, courseId, quizAnswers)

Input: user (Object), courseId (String), quizAnswers (Object)

Output: result (Object) or error (String)

Pseudocode:

1. Validate user enrollment in the course.
2. Retrieve correct answers from the database.
3. Compare submitted answers with correct answers.
4. Calculate score and store result in database.
5. Return the result object with score and status.

5.1.5 In-App Chat with Admin

Function: sendMessage(user, message)

Input: user (Object), message (String)

Output: success (Boolean) or error (String)

Pseudocode:

1. Check if user is authenticated.
2. Store message with timestamp under the chat thread with admin.
3. Optionally notify admin.
4. Return success.

Function: receiveMessages(user)

Input: user (Object)

Output: messageList (List<Message>)

Pseudocode:

1. Retrieve chat thread between user and admin.
2. Return list of messages.

5.1.6 Profile Management

Function: updateProfile(user, updatedFields)

Input: user (Object), updatedFields (Map<String, String>)

Output: success (Boolean) or error (String)

Pseudocode:

1. Validate user session.
2. Update provided fields (e.g., username, email, phone) in the database.
3. Return success.

5.1.7 Generate Personalized Learning Roadmap

Function: generateRoadmap(user, skills)

Input: user (Object), skills (Array of Strings)

Output: roadmap (Object) or error (String)

Pseudocode:

1. Validate user authentication and input skills.
2. If skills list is empty, return an error message.
3. Retrieve relevant learning paths based on entered skills from the database.
4. Analyze user skill level and categorize topics into Beginner, Intermediate, and Advanced.
5. Generate a structured roadmap with ordered topics for each level.
6. Store the generated roadmap in the database for the user.

7. Return the roadmap object containing all levels and recommended topics.

5.1.8 RAG-Based Video Assistant Query Handling

Function: askVideoAssistant(user, videoId, query)

Input: user (Object), videoId (String), query (String)

Output: response (String) or error (String)

Pseudocode:

1. Validate user authentication and access to the video.
2. Validate that the query is not empty.
3. Retrieve video transcript and metadata from the database.
4. Preprocess the query and convert it into embeddings.
5. Search relevant chunks from the transcript using similarity matching.
6. Pass the retrieved context and query to the RAG model.
7. Generate a contextual response based on retrieved information.
8. Log the query and response for future improvement.
9. Return the generated response to the user.

5.1.9 Book Live Session (SMeet)

Function: bookLiveSession(user, instructorId, sessionRequest)

Input: user (Object), instructorId (String), sessionRequest (Object)

Output: bookingConfirmation (Object) or error (String)

Pseudocode:

1. Validate user authentication and session request details.
2. Check if the selected instructor exists and is available.
3. Validate requested date and time slot.
4. Check for scheduling conflicts for both user and instructor.
5. If the slot is unavailable, return an error message.
6. Reserve the time slot temporarily.
7. Create a live session booking entry in the database.
8. Generate session link and session details.
9. Send notification to both user and instructor.

10. Confirm the booking and update slot availability.
11. Return booking confirmation with session details.

5.2 User Authentication

The Skill Forge app integrates Firebase Authentication to ensure a secure and reliable login system for users. Upon successful sign-up or login, Firebase generates a unique user ID and assigns appropriate roles such as student, instructor, or admin.

These credentials are securely encoded in authentication tokens that can be used across the platform. This role-based access control ensures that each user interacts only with the functionalities appropriate to their profile, thereby minimizing unauthorized access and maintaining the platform's integrity.

Using token-based authentication rather than traditional session tracking allows Skill Forge to scale efficiently across devices and platforms. Each token is independently verifiable, which eliminates the need to maintain server-side session state.

This approach not only enhances performance and responsiveness but also strengthens security by reducing vulnerability to session hijacking and cross-site scripting attacks. Furthermore, it simplifies the process of integrating new devices or expanding platform features without compromising security standards.

5.3 User Registration Algorithm

The User Registration Algorithm manages the complete process of registering new users onto the Skill Forge platform. Whether a user is a student, instructor, or administrator, this algorithm ensures a smooth and user-friendly enrollment experience.

When a new user fills out the registration form with necessary details such as name, email, password, and role, the algorithm validates the inputs for correctness and completeness. It checks for duplicates (like existing emails), enforces password strength rules, and ensures the selected role aligns with the platform's access control policies.

Once the input is validated, the algorithm securely stores user data in the Firebase database using encryption and best practices to protect sensitive information. Additionally, the system generates a request notification for the admin, who can then review the account details and either approve or reject the registration. This two-step verification helps maintain the integrity of the platform by preventing unauthorized access and ensuring only verified individuals can use their respective roles. This approach supports scalability, security, and organized user management across the platform.

5.4 Course Enrollment and Progress Tracking Algorithm

The Course Enrollment and Progress Tracking Algorithm is a core component of the Skill Forge platform, designed to facilitate personalized learning experiences. When a student browses available courses and selects one to enroll in, this algorithm manages the enrollment process by validating the user's eligibility.

And updating the enrollment database, and granting access to course materials such as videos, quizzes, and assignments. Upon successful enrollment, a course-specific progress tracker is initialized for that student, allowing the platform to monitor each learner's journey individually.

As the student engages with course content watching lectures, attempting quizzes, or submitting assignments the system records every interaction in real time. This progress is visualized using percentage-based indicators and milestone completions, helping students understand how far they've come and what remains.

The algorithm also enables features like resuming where the student left off, generating performance reports, and providing personalized recommendations based on progress. Overall, it ensures a seamless, interactive, and data-driven learning experience that empowers users to stay on track and motivated.

5.5 Security and Data Protection

Security is a top priority within the Skill Forge app, ensuring that user trust and data integrity are never compromised. The platform employs advanced encryption protocols to safeguard sensitive user information, including personal details, course materials, authentication tokens, and chat logs.

All data transmitted between the user's device and the server is encrypted using secure communication protocols, protecting it from potential interception or unauthorized access. To further strengthen data protection, Skill Forge implements secure cloud storage practices and utilizes API rate limiting to prevent misuse or excessive access attempts that could compromise the system.

In addition to standard encryption, Skill Forge incorporates device-level security features to protect content integrity and discourage unauthorized distribution. Users are restricted from taking screenshots or initiating screen recordings while engaging with the app, particularly during video playback or when viewing protected materials.

This measure helps prevent content piracy and ensures that course materials remain exclusive to authorized users only. By combining application-layer security with hardware-level restrictions, Skill Forge ensures a safe, privacy-respecting environment for all learners and administrators.

The system also enforces strong authentication mechanisms to ensure that only authorized users can access their accounts. Regular security updates and monitoring are performed to detect and address potential vulnerabilities. Role-based access control is implemented to limit access to sensitive data based on user permissions. Additionally, user sessions are managed securely to prevent unauthorized access from inactive or compromised sessions. These measures collectively enhance the overall security and reliability of the platform.

Table 5.1 External API

Name of API	Description of APIs	Purpose of Usage	Function/Class Name
Firebase Authentication	Allows secure and easy user authentication using email/password or other methods.	Handles login, sign-up, and authentication for all users (student, admin, head).	Login Activity, Register Activity, Auth Service
Firebase Firestore	A cloud-based NoSQL database that stores and syncs app data in real time.	Stores data like events, societies, feeds, and user profiles.	Event Repository, Society Service, Feed Adapter

5.6 Roadmap Generation Algorithm

The Roadmap Generation Algorithm handles the process of creating a personalized learning path for users on the Skill Forge platform. When a user enters the skills they already possess and clicks the generate button, the system first validates the input to ensure that the provided skills are not empty and are in a usable format. It then retrieves relevant learning paths and topic data from the database based on the entered skills.

The algorithm analyzes the user's current skill set and maps it against predefined learning structures to identify gaps and progression levels. Based on this analysis, it organizes the roadmap into three clear stages: Beginner, Intermediate, and Advanced. Each stage contains ordered topics that guide the user step-by-step toward mastery. Once the roadmap is generated, it is stored in the database for future access, allowing users to revisit and track their progress. This approach ensures a structured, personalized, and scalable learning experience for every user.

5.7 RAG-Based Video Assistant Algorithm

The RAG-Based Video Assistant Algorithm manages how users interact with video content by allowing them to ask questions and receive contextual answers. When a user submits a query related to a specific video, the system first validates the user's access and ensures the query is not empty. It then retrieves the video transcript and associated metadata from the database.

The algorithm processes the user's query and converts it into embeddings to enable semantic understanding. It performs similarity matching to identify the most relevant segments of the video transcript. These relevant chunks are then passed, along with the original query, to the RAG (Retrieval-Augmented Generation) model. The model generates a response based only on the retrieved context, ensuring accuracy and relevance to the video content. The response is then returned to the user, and the interaction may be logged for system improvement. This method provides precise, context-aware assistance without relying on unrelated information.

5.8 SMeet Live Session Booking Algorithm

The SMeet Live Session Booking Algorithm manages the process of scheduling live interactive sessions between users and instructors. When a user requests to book a session, the system first validates the user's authentication and ensures that all required booking details, such as instructor selection and preferred time, are provided. It then checks whether the selected instructor exists and is available during the requested time slot.

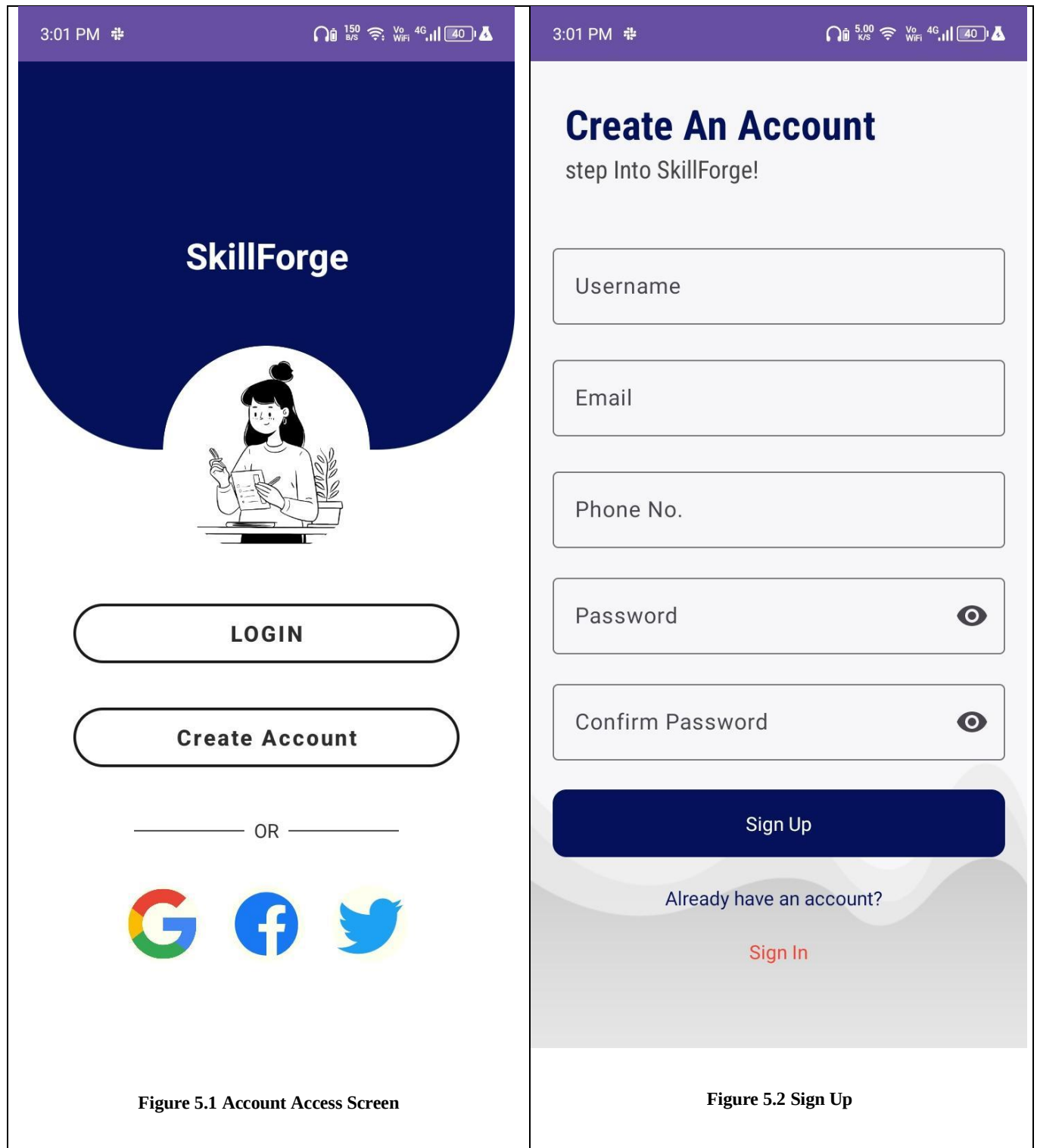
The algorithm verifies that there are no scheduling conflicts for either the user or the instructor. If the requested slot is unavailable, the system returns an appropriate error message. If available, the slot is temporarily reserved, and a booking entry is created in the database. The system then generates session details, including a unique session link, and sends notifications to both the user and the instructor. Finally, the booking is confirmed, and the time slot is marked as occupied. This process ensures efficient scheduling, avoids conflicts, and enables smooth real-time interaction between users and instructors.[\[8\]](#)

5.9 Take Quiz

The Quiz Attempt and Result Evaluation Algorithm manages the process of assessing a user's understanding after completing a course on the Skill Forge platform. Before attempting the quiz, the system ensures that the user has successfully completed the required course video. This validation step guarantees that only eligible users can proceed to the assessment. Once verified, the user can attempt the quiz by submitting their answers.

5.10 User Interface

The following figures represent the user interfaces and screens of the Skill Forge e-learning app:



The Fig 5.1 provides the user with multiple options to access their account. The user can log in manually

by entering their registered email or username along with their password into the provided input fields. Alternatively, the user can choose to log in using external authentication providers, including options such as Google or Facebook, eliminating the need to enter credentials manually. A divider labeled "OR" visually separates the manual login method from the social login options. For new users who do not yet have an account, a "Create Account" button is available to navigate to the registration screen. This screen functions as the primary authentication gateway, giving users flexibility in how they sign in. If the user already has an account, they can simply log in using their existing credentials without needing to register again.

The Fig 5.2 allows a new user to register for the application by entering the required information into several fields. The user must provide a username, email address, phone number, and a password, followed by confirming the password to ensure accuracy. Once all fields are filled correctly, the user submits the registration by clicking the "Sign Up" button. If the user already has an existing account, there is a "Sign In" button available on this screen, which allows them to navigate back to the login screen and use their previous credentials instead of completing the registration process. This functionality ensures that returning users are not forced to create a duplicate account. The screen effectively separates new user registration from existing user login while maintaining a straightforward path for both scenarios.

This Fig 5.3 screen allows the user to log into the application using their existing credentials. The user must enter a username and a password into the two respective input fields. Below the password field, a "Forgot Password?" option is provided, enabling the user to initiate a password recovery process if they cannot remember their credentials. Once the username and password are entered, the user clicks the "LOGIN" button to authenticate and access their account. For users who do not yet have an account, a text link labeled "Not yet registered? SignUp Now" is available at the bottom of the screen. Clicking this link navigates the user to the registration screen. This screen serves as the standard authentication method without any social login options, focusing solely on manual credential-based access.

This Fig 5.4 displays the profile details of the currently logged-in user. The top of the screen shows the application name "Skill Forge". The user's registered information is displayed in plain text, including an email address, the same email address repeated, and a phone number. Below this displayed information, the screen provides five actionable menu options for the user to manage their account. The user can select "Edit Profile" to modify their personal information, "Change Password" to update their login credentials, or "Delete User" to permanently remove their account from the application. Additionally, "About Us" and "Privacy Policy" options are available to provide the user with information about the application and its data handling practices. This screen functions as a central hub for account management and application

information.

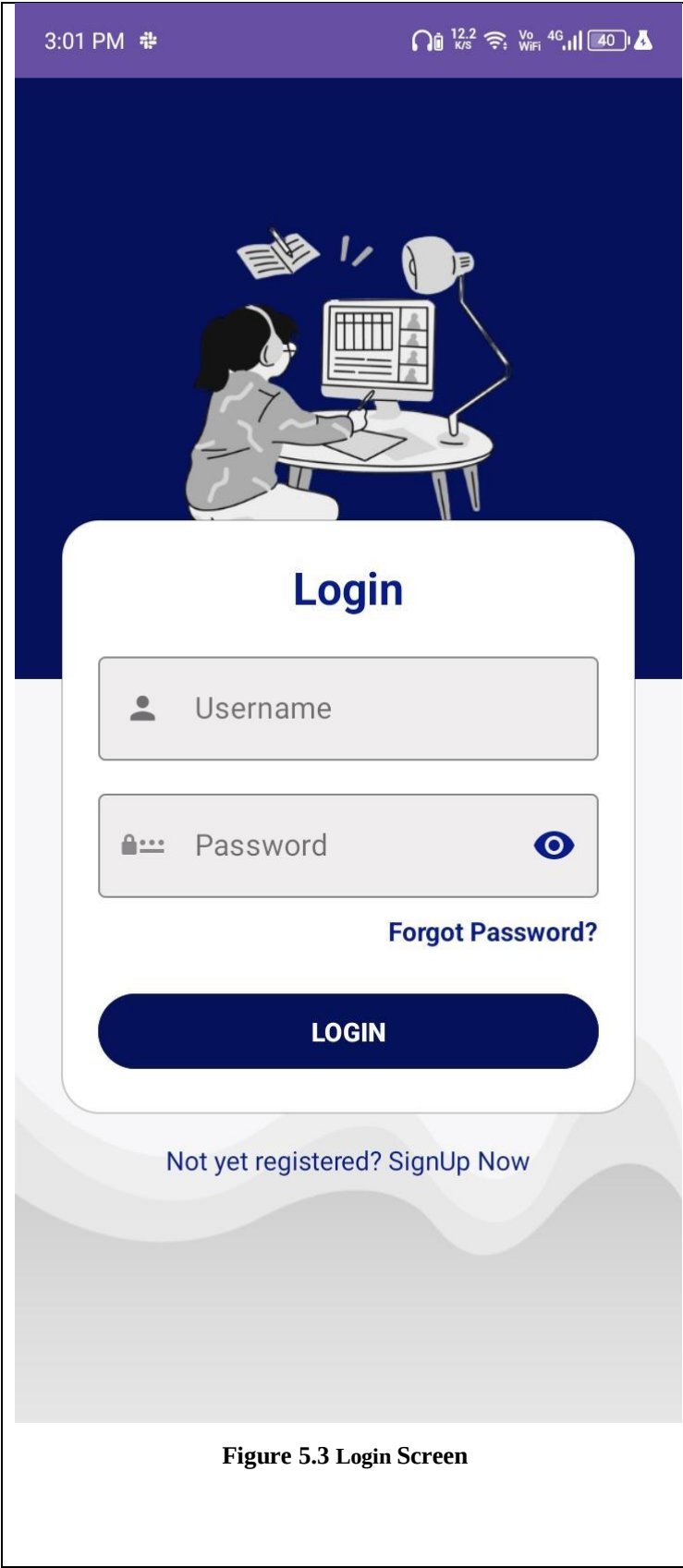


Figure 5.3 Login Screen

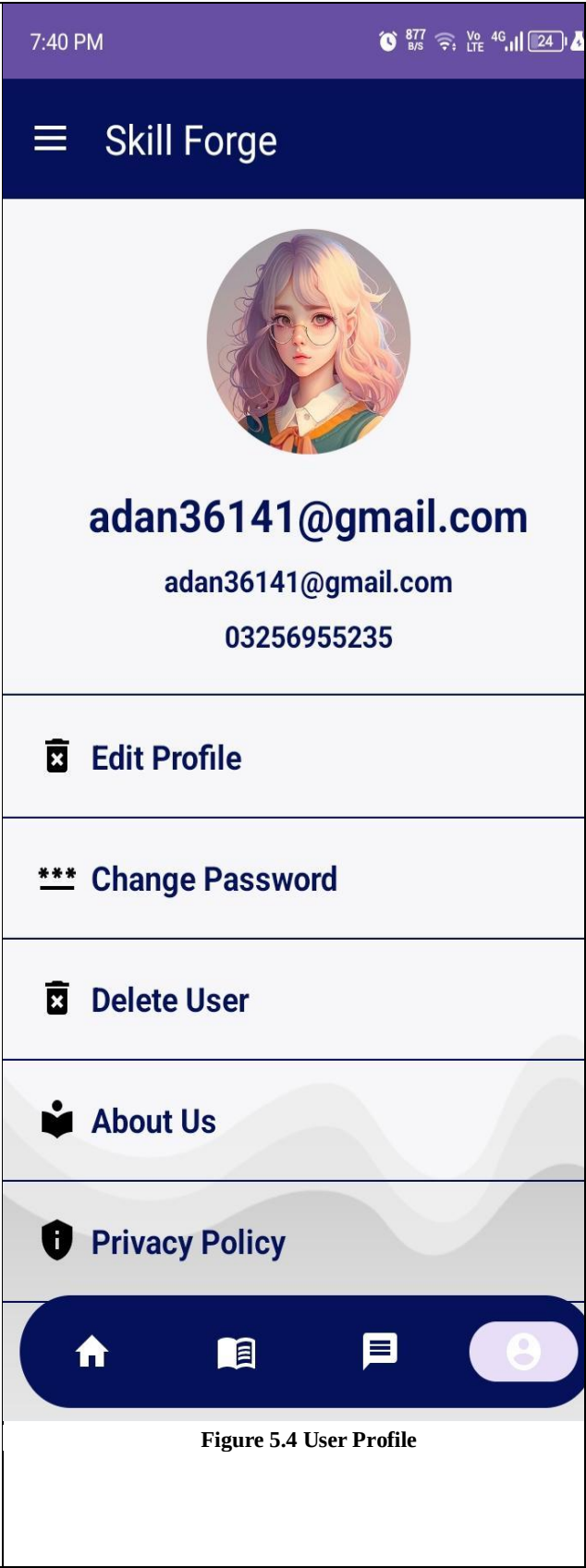


Figure 5.4 User Profile

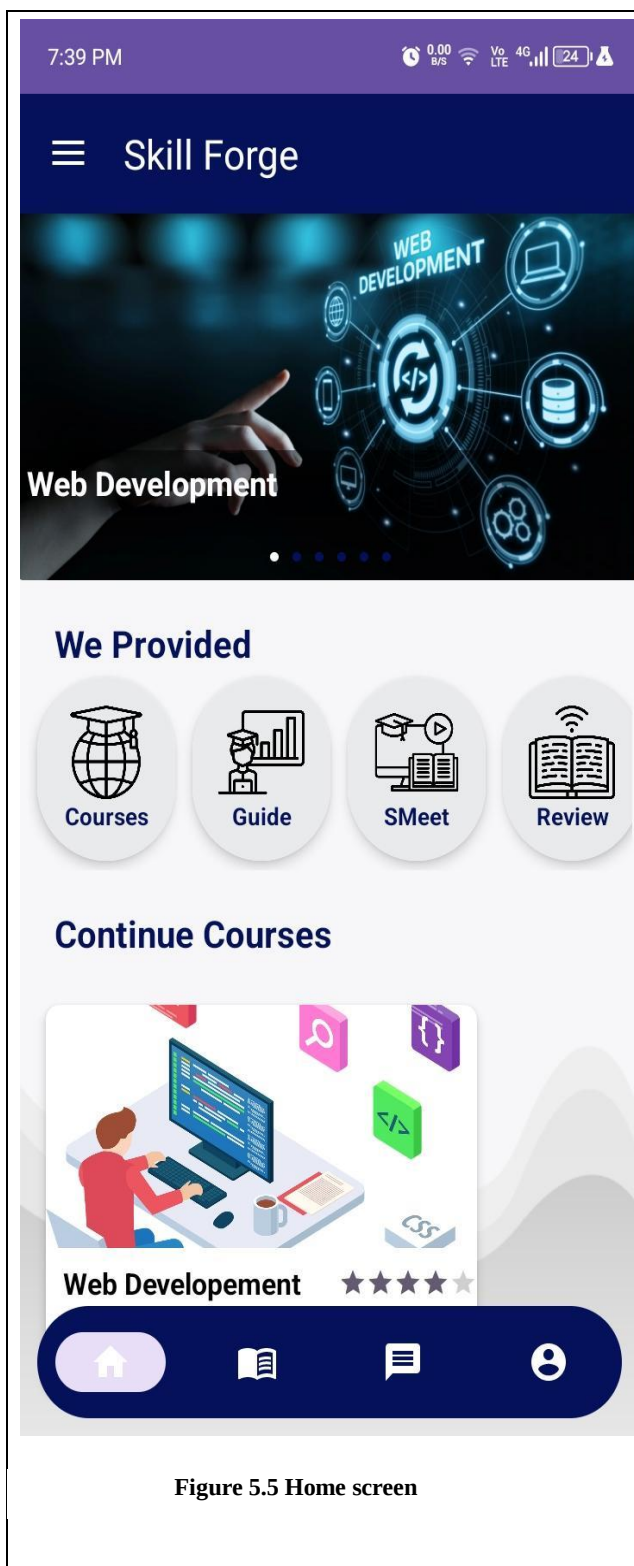


Figure 5.5 Home screen

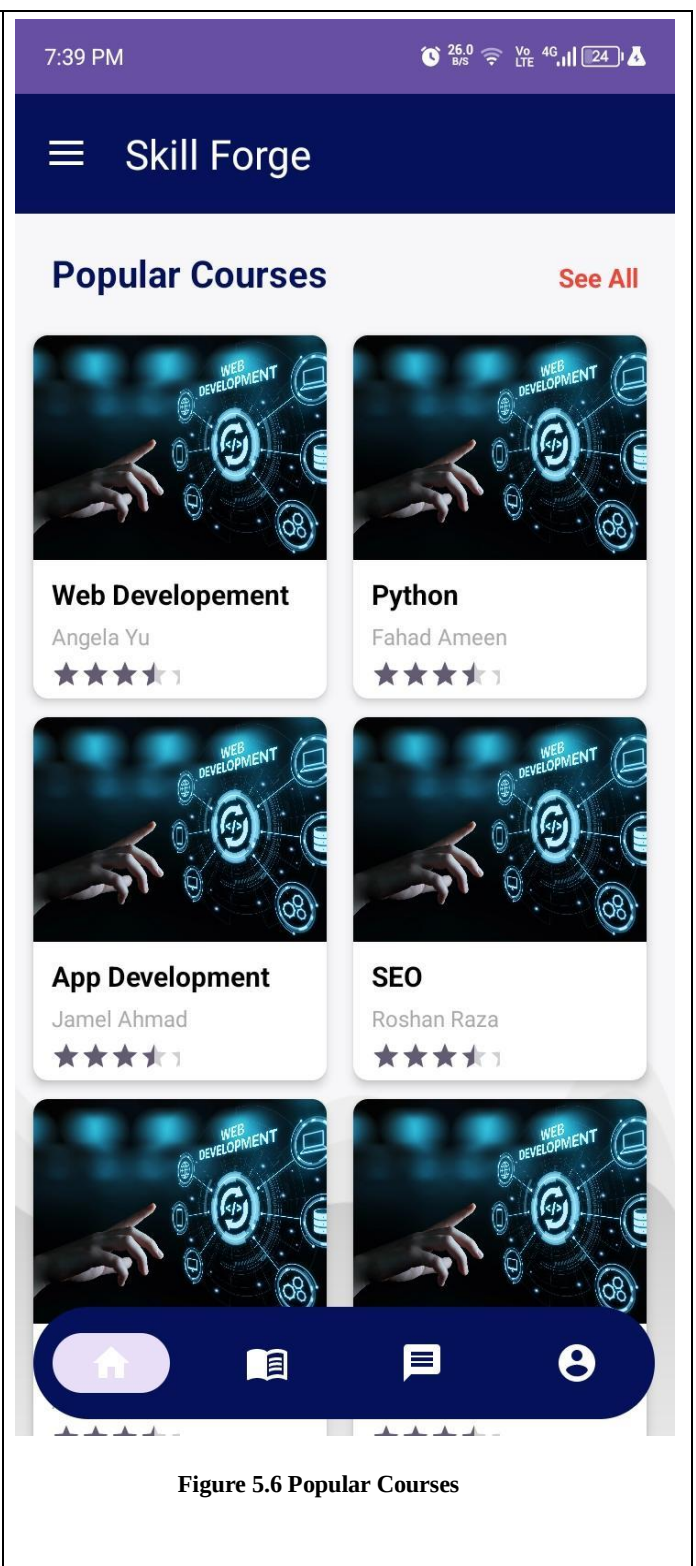


Figure 5.6 Popular Courses

The Fig 5.5 shows a mobile application interface named “Skill Forge” displayed at the top of the screen with a menu icon on the left side. Below the header, a large banner section presents the title “Web Development” along with a digital technology-themed image. Under the banner, small dots indicate that the screen may contain multiple sliding banners. The next section is titled “We Provided,” where four circular icons are arranged horizontally. These icons are labeled Courses, Guide, Smeet, and Review to

represent different features of the application. Below this, the “Continue Courses” section displays a course card for Web Development with an illustration and a star rating. At the bottom of the interface, a navigation bar contains four icons that provide access to different sections of the application.

This Fig 5.6 presents the main dashboard of the Skill Forge application, focused on showcasing available courses. At the top, the app title is displayed alongside a menu icon for navigation. Below it, a “Popular Courses” section highlights featured learning options. A “See All” option allows users to explore the complete course catalog. The courses are displayed in a clean card-based grid layout. Each card includes an image, course title, and instructor name. User ratings are shown beneath each course to indicate quality and feedback. The layout is visually organized, making it easy to browse multiple courses at once. The consistent design of cards helps maintain a uniform look across the screen. At the bottom, a navigation bar provides quick access to key sections of the app. Overall, the interface is user-

friendly and designed for smooth course discovery.

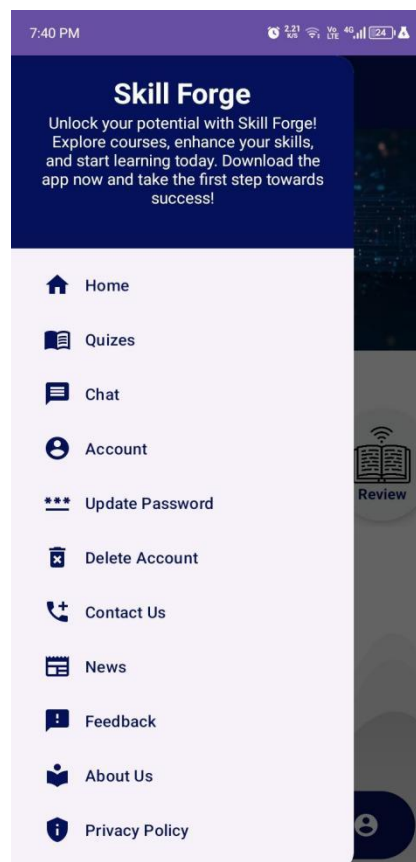


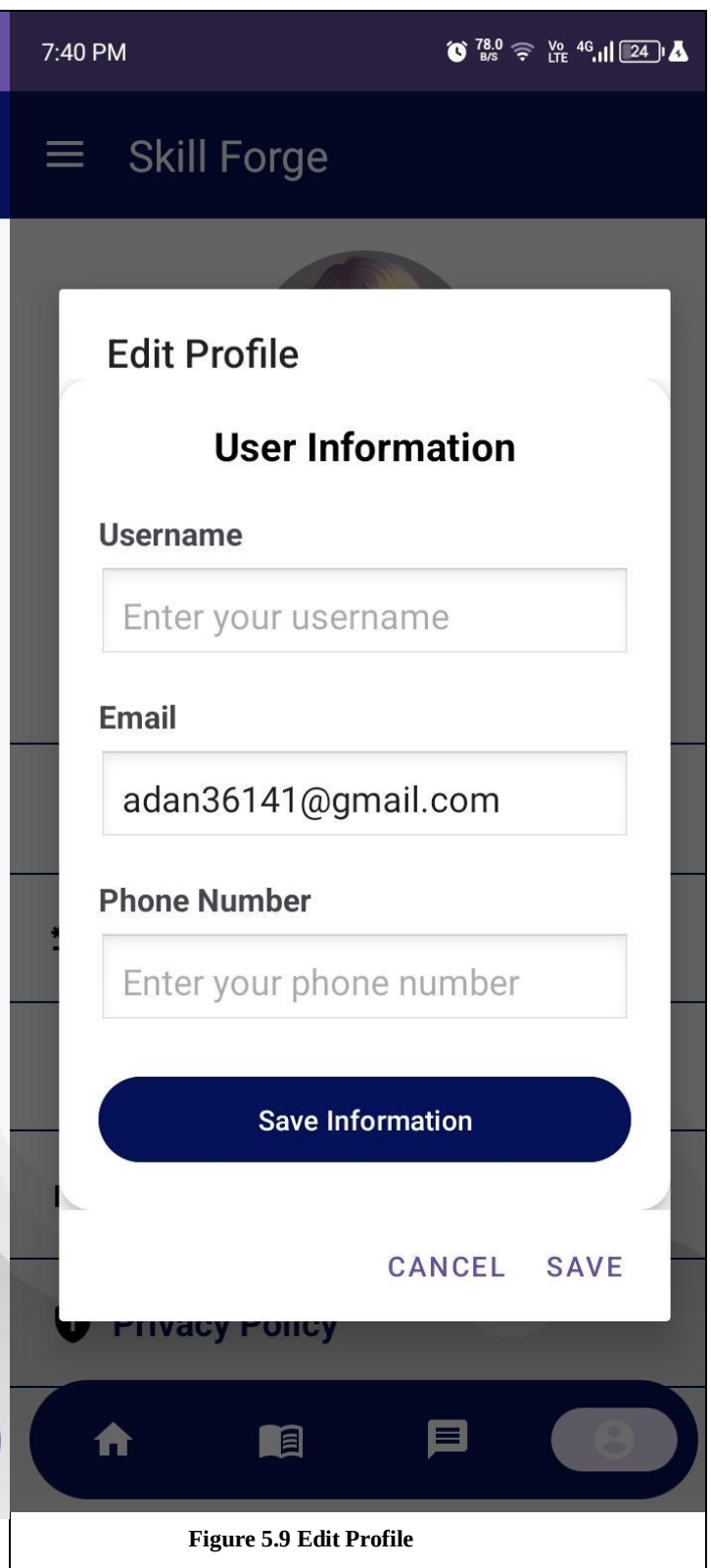
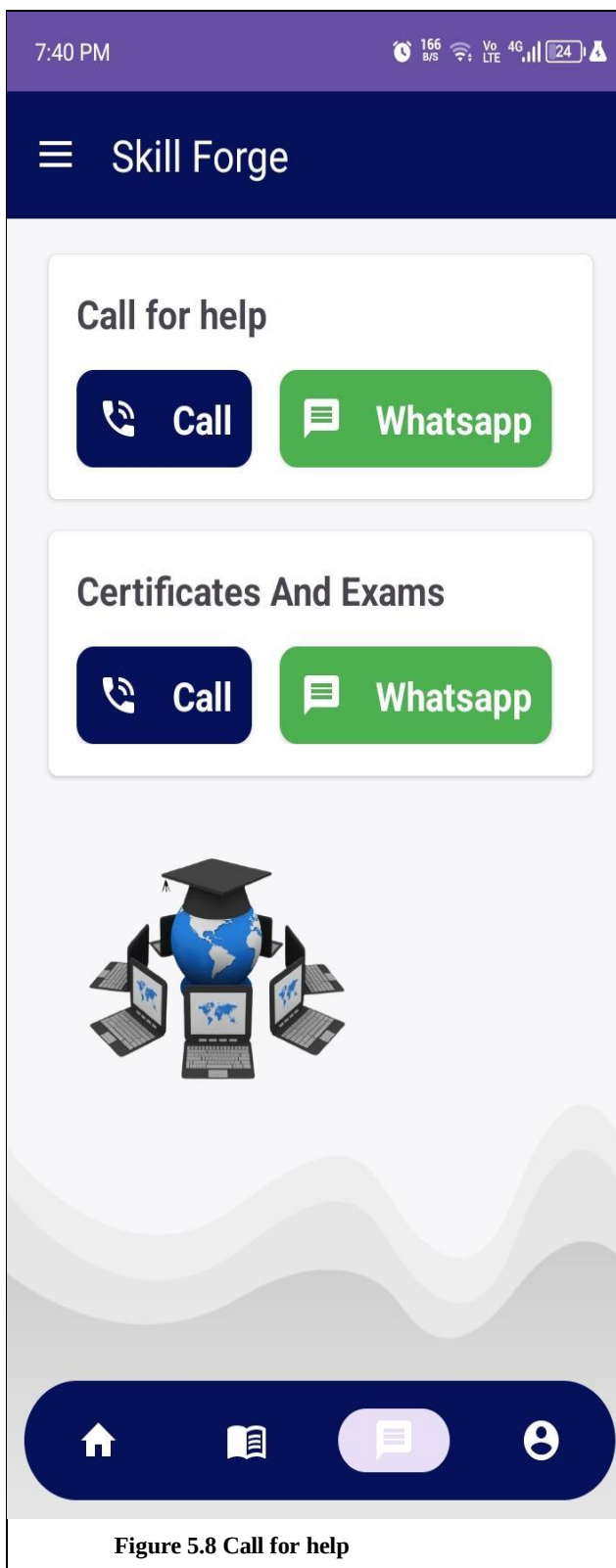
Figure 5.7 Side Bar

The Fig 5.7 shows the side navigation menu of the Skill Forge mobile application. At the top of the menu, the application name is displayed in bold with a short description that encourages users to unlock their

potential through skill development and online courses. Below the header, a vertical list of navigation options is presented with icons beside each item. The menu includes options such as Home, Quizzes, Chat, Account, Update Password, Delete Account, Contact Us, News, Feedback, About Us, and Privacy Policy. The menu appears as a white panel sliding from the left side of the screen, partially covering the main interface in the background. Each option is arranged clearly with proper spacing to make the menu simple and user-friendly for navigation.

The Fig 5.8 shows a Skill Forge mobile application screen that provides direct communication support for users. At the top, the application header displays the app name along with a menu icon. Below the header, two support sections are shown in separate cards labeled “Call for help” and “Certificates And Exams.” Each card contains two action buttons that allow users to either make a phone call or send a message through WhatsApp. This integration enables users to quickly contact the support team for assistance or inquiries without leaving the application. A centered educational image is displayed below the contact sections, while the bottom navigation bar provides quick access to other features of the application.

The Fig 5.9 shows the Edit Profile screen of the Skill Forge mobile application. A centered pop-up form appears over the main interface with the heading “User Information.” The form contains input fields for Username, Email, and Phone Number to allow users to update their personal details. The email field already displays an existing email address, while the other fields provide placeholder text for user input. Below the fields, a “Save Information” button is provided to submit the updated profile details. At the bottom of the pop-up, Cancel and Save options allow users to close the form or confirm the changes. This screen enables users to manage and update their account information directly within the application.



The **Skill Forge** application presents a clean and user-friendly interface that supports a smooth learning experience for its users. The design uses a consistent color scheme, clear typography, and organized

spacing that make the content easy to read and understand. Navigation throughout the application appears simple because important features are arranged in a structured manner, allowing users to move between sections without confusion. The layout of each screen is visually balanced, helping users quickly identify available actions and access key functions with minimal effort. Interactive elements such as buttons, icons, and menus are placed in accessible positions, which improves usability on mobile devices. The interface also maintains consistency across different screens, creating a connected experience that feels familiar as users continue using the application. Support features integrated within the app reduce the need to switch between platforms, which contributes to a more convenient experience. Profile management and course-related sections are presented in a straightforward way that reduces complexity for the user. The overall user interface and user experience reflect a practical design approach that focuses on clarity, accessibility, and ease of interaction for learners.

Chapter 6

Testing and Evaluation

6. Testing

6.1 Unit Testing

Unit tests is conducted to assess the individual components of the Skill Forge.

Table 6.1 Login as a Student

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Student login with valid credentials	Username: <u>lawa01@gmail.com</u> Password: 12345678 Username: <u>adan773@gmail.com</u> Password: 12345678	Student successfully logs in to the system	Pass

The unit test documented in Table UT-001: Student Login with Valid Credentials is designed to verify the functionality of the student login feature within the Skill Forge platform. In this test, two different sets of valid student login credentials are used: lawa01@gmail.com / 12345678

and adan773@gmail.com / 12345678. The test aims to ensure that the system correctly authenticates registered students and grants them access to their respective accounts.

The expected result is that both users should be able to log in successfully without errors. Upon executing the test, the system responded as expected authenticating both students and allowing them to access the platform.

Since the actual result matched the expected behavior, the test case is marked as Pass, confirming that the student login functionality is working reliably with valid credentials.

Table 6.2 Logout as Student

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Click the logout button and logout	Logout	Student Successfully exit from the app	Pass

Logout as a Student presents a unit test designed to validate the logout functionality for student users on the Skill Forge platform. The test case with ID UT-001 assesses whether the system allows a student to successfully exit the application after clicking the logout button.

The attribute tested is the logout operation itself, where the student initiates the action to log out of their session. The expected result is that the student should be securely and completely logged out of the app.

The actual result confirmed that the logout process worked as intended, and the student was exited from the application without any issues. As the output aligned with expectations, the test was marked as Pass, confirming that the logout functionality for student users operates reliably.

Table 6.3 Unit Testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	User Registration	Valid user information	User account is created successfully	Pass
UT-002	User Authentication	Valid user credentials	Successful authentication and user login	Pass
UT-003	Password Field	Leave the password field blank during registration	System displays an error message	Pass
UT-004	Course Enrollment	Enrolling in a course	User is successfully enrolled in the course	Pass
UT-005	Quiz Submission	Submitting a quiz after completing it	Quiz is submitted successfully, and results are shown	Pass
UT-006	Admin Course Management	Admin adds a new course	Course is successfully added to the platform	Pass

Unit Testing outlines a comprehensive evaluation of critical functionalities within the Skill Forge e-learning platform. Each test case verifies that specific user actions and system processes function correctly and meet the expected outcomes.

The first test case, UT-001 (User Registration), checks whether the platform successfully creates a new account when valid user information is provided. The system performed as expected, confirming successful account creation, and the test passed.

In UT-002 (User Authentication), the test validates that a user with correct credentials can log in

without issues. The successful authentication confirmed the login mechanism's reliability, resulting in a pass.

UT-003 tests system behavior when a user attempts registration with the password field left blank. As expected, the system displays an error message prompting the user to enter a password, ensuring input validation is functioning correctly.

In UT-004 (Course Enrollment), the system is tested for enrolling a user into a course. The user is successfully enrolled, verifying the enrollment logic works effectively.

The UT-005 (Quiz Submission) test involves submitting a completed quiz. The system accepts the submission and displays results promptly, confirming the quiz functionality is accurate and responsive.

Lastly, UT-006 (Admin Course Management) checks whether an admin can add a new course to the platform.

6.2 Manual Testing

During the manual testing phase, we rigorously assessed "Skill Forge" for various functionalities. Any issues identified during testing were promptly addressed to guarantee a seamless user experience within the application.

Table 6.4 : Manual Testing

Test Cases	Expected Result	Actual Result	Pass/Fail
App Launch	Successful	Successful	Pass
User Registration	Successful	Successful	Pass
User Login	Successful	Successful	Pass
Home Page Display	Accurate	Accurate	Pass
Search Course	Accurate	Accurate	Pass
Enroll in Course	Successful	Successful	Pass
Quiz Attempt and Submission	Successful	Successful	Pass
Manage Profile	Successful	Successful	Pass
App Performance and Response	Smooth	Smooth	Pass
App Navigation	Easy	Easy	Pass
App Usability	High	High	Pass
Logout	Successful	Successful	Pass

Manual Testing provides an overview of the testing results for various functional aspects of the Skill Forge app. Each test case is designed to ensure that critical features and functionalities of the platform work as intended in real-world scenarios.

The App Launch test confirms that the application launches successfully, which passed without issues. The User Registration and User Login tests confirm that new users can register and log in smoothly, both of which also passed, ensuring the authentication system is functioning correctly.

The Home Page Display test checks the accuracy of the content presented to the user on the home screen, and it passed, confirming that users are presented with the correct information upon accessing the app.

For the Search Course test, it was verified that the search feature correctly returns accurate results when users search for courses, which passed. The Enroll in Course test ensures users can enroll in a course without complications, and this also passed, confirming seamless course enrollment.

The Quiz Attempt and Submission test confirms that quizzes can be attempted and submitted successfully, which passed, ensuring the platform's assessment feature works as expected.

The App Navigation test checks whether users can easily navigate through the app, and this passed, confirming an intuitive user interface. App Usability and User Friendliness ensures that the app provides a user-friendly experience, and it also passed, showing that the app is designed with ease of use in mind. Finally, the Logout test confirms that users can successfully log out of the app, which passed, completing the set of tests for critical app functions. All tests in the table passed, verifying that the Skill Forge app functions as intended and provides a positive user experience.

6.3 Functional Testing

Functional testing verifies End to end functionalities of a Skill Forge against its functional requirements and specifications, to ensure that the software behaves as expected.

Functional testing focuses on validating each feature of the system by providing appropriate inputs and examining the outputs against expected results. It ensures that all modules, such as user registration, course browsing, video playback, chat system, and certification requests, operate correctly. Test cases are designed based on functional requirements to cover both typical and edge-case scenarios. This type of testing helps identify issues in business logic, data handling, and user interactions. It also ensures that integrations between different components function smoothly. Errors detected during functional testing

are documented and resolved before deployment. Overall, it plays a crucial role in delivering a reliable and user-focused application.

Table 6.5 Functional Testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
FT-001	User Registration	Registering with valid university email address	User accounts can be created with valid email addresses	Pass
Ft-002	Searching Course	Searching for a course	If the course exists, results should be displayed	Pass
FT-003	Real-Time Notifications	Sending and receiving notifications	Notifications are sent and received in real-time	Pass
FT-004	Course Enrollment	Enrolling in a course	Users can enroll in courses successfully	Pass
FT-005	User Profile Management	Editing and updating user profiles	User profiles can be edited and updated	Pass
FT-006	Password Encryption	Password encryption	Passwords are securely encrypted a	Pass

Functional Testing presents a detailed overview of the functional tests conducted on various features of the Skill Forge app. Each test is designed to validate the core functionalities and ensure the system operates as expected in real-world scenarios.

The User Registration test (FT-001) checks if users can successfully create accounts using a valid university email address, which passed, confirming that the registration process functions correctly with authorized email addresses.

The Searching Course test (FT-002) verifies that users can search for courses, and if the course exists, the system correctly displays the results. This test also passed, indicating the search functionality works as intended.

The Real-Time Notifications test (FT-003) evaluates whether notifications are sent and received promptly. This test passed, confirming that the app supports real-time communication, keeping users

informed of updates and activities in the app.

The Course Enrollment test (FT-004) ensures that users can successfully enroll in courses. This functionality passed as well, verifying that the course enrollment system is working seamlessly.

The User Profile Management test (FT-005) assesses whether users can update and edit their profiles as needed, such as changing personal information or preferences. This test also passed, confirming that profile management features are functional and user-friendly.

Finally, the Password Encryption test (FT-006) checks whether user passwords are securely encrypted. The test passed, confirming that the app employs secure password encryption to protect user data from unauthorized access.

Overall, all functional tests in this table have passed, demonstrating that the Skill Forge app's critical features, including registration, search, notifications, course enrollment, profile management, and password security, are operating as expected.

6.4 Integration Testing

Integration testing is a software testing technique that focuses on evaluating the interactions between different software modules or components of Skill Forge.

Table 6.6 Integration testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
IT-001	Test and configure Firebase	Firebase real-time connection	Connection Established	Pass
IT-002	Test and configure the database.	Database real-time connection	Connection established	Pass

Integration Testing outlines two key tests conducted to validate the integration of different system components in the Skill Forge app. These tests focus on ensuring that the communication between modules, such as web socket connections and the database, functions properly when integrated.

The first test, Test and configure Web Socket (IT-001), verifies the real-time communication functionality of the app through web sockets. Firebase are used to maintain continuous

communication between the client and server, ensuring real-time updates and interactions within the app. The expected result is the successful establishment of a web socket connection, which was achieved, as indicated by the "Connection Established" outcome. This confirms that the app can reliably use web sockets for real-time communication, such as notifications or messaging between users and admins.

The second test, Test and configure the Database (IT-002), checks the real-time connection between the app and the database. This is essential for ensuring that the app can retrieve and store user data, course content, and other relevant information in real time without delays or inconsistencies.

The expected result of this test is that a connection is successfully established with the database, which was also achieved, as indicated by the "Connection established" outcome. This confirms that the database integration is functioning correctly, allowing the app to maintain up-to-date and accurate data.

6.5 Tools and Technologies

Table 6.7 Tools and Technologies

Tools And Technologies	Tools	Version	Rationale
	Android Studio	2023.2.1	Kotlin based project
	Firebase Server	2015	DBMS
	MS Excel	2016	Gantt Chart
	Android Studio	2023.2.1	Making virtual devices
	MS Word	2015	Documentation
	MS Power Point	2015	Presentation
	Pencil	2.0.5	Mockups Creation
	Technology	Version	Rationale
	Kotlin	1.9.20	Programming language
	Android		Platform
	Non-SQL		Firebase
	Html	5	Admin

Tools and Techniques provides a comprehensive list of tools, technologies, and their versions used in the development of the Skill Forge app. These tools and technologies are essential for various phases of the project, such as development, documentation, testing, and presentation. Each tool serves a specific purpose that contributes to the overall efficiency and effectiveness of the development process.

Android Studio 2023.2.1 is used as the primary integrated development environment (IDE) for building the Skill Forge app. Since the app is developed using Kotlin, this version of Android Studio supports Kotlin-based projects and provides essential features such as code completion, debugging, and testing tools.

It also enables the creation of virtual devices for testing the app on different Android configurations. Firebase Server 2015 is used as the database management system (DBMS) for storing and managing the app's data. Firebase provides a scalable, real-time NoSQL database that facilitates secure data storage and efficient data retrieval.

For project management and documentation, MS Excel 2016 is utilized to create Gantt charts, which help in planning and tracking project milestones, tasks, and timelines.

MS Word 2015 is employed for documentation purposes, including writing reports and user manuals, while MS PowerPoint 2015 is used for creating presentations.

Pencil 2.0.5 is used for creating mockups and wireframes, which visually represent the app's user interface design before development.

The app's core programming language is Kotlin 1.9.20, which is known for its concise syntax, interoperability with Java, and compatibility with Android development. The app is built on the Android platform, which provides the necessary tools and frameworks for developing Android applications.

For the admin panel, HTML5 is used for building the web-based interface that administrators will use to manage the platform. Firebase is employed as the Non-SQL database to handle real-time data synchronization and secure user authentication.

Additionally, for the basic admin panel, the project utilizes Django and Python. Django is a high-level Python web framework that helps in building robust and scalable web applications quickly. It provides built-in features like authentication, routing, and admin interfaces, making it ideal for managing user data, content, and various administrative functions efficiently.

Chapter 7

Conclusion and Future Work

7. Conclusion and Future Work

7.1 Conclusion

The Skill Forge e-learning app stands as a groundbreaking development in the realm of modern education, offering a comprehensive and dynamic platform that caters to the needs of students, instructors, and administrators alike. By leveraging advanced technology and real-time functionalities, the app has addressed many of the challenges traditionally faced by educational systems, such as access to learning materials, real-time communication, and progress tracking.

The app's user-friendly interface, coupled with features designed to foster interactivity, has transformed the learning experience for students and educators, making it both engaging and efficient.

A key aspect of the Skill Forge app is its focus on creating a connected and collaborative learning environment. Through features such as course progress tracking, real-time messaging, and peer interaction, students are empowered to take charge of their learning.

They can monitor their academic achievements, access relevant resources, and engage with instructors and fellow students seamlessly. These capabilities not only enhance students' academic performance but also promote a sense of community and interaction within the learning environment.

Furthermore, the app enables instructors and academic staff to manage their courses, assignments, and other resources more efficiently, streamlining administrative tasks and fostering a more organized academic environment.

Admins can easily manage these opportunities, adding new postings, editing existing ones, or removing outdated listings, ensuring that students have access to the latest and most relevant employment opportunities in their field of study.

This module also supports a holistic approach to career development, complementing the educational experience by giving students access to practical opportunities that enhance their learning and skillset.

As we conclude this project, it is clear that Skill Forge has successfully met its objectives of creating an effective, adaptable, and user-centric e-learning solution. The app's focus on user

experience, seamless navigation, and real-time interaction has led to a significant improvement in both student engagement and educator efficiency. The Skill Forge app is a testament to the future of education, where learning is personalized, interactive, and closely linked to real-world applications.

7.2 Future Work

While the Skill Forge e-learning app has successfully achieved its primary goals, there are always opportunities for growth and refinement in the pursuit of creating an even more comprehensive and user-centered learning platform. Several key areas of focus for future development will help improve the app's functionality, scalability, and overall user experience:

7.2.1 Job and Internship Search Portal

A valuable future enhancement for the Skill Forge app would be the integration of a job and internship search portal. This feature would connect students with relevant career opportunities based on their skills, course progress, and performance within the app. By analyzing completed courses, certifications, and learning outcomes, the system could recommend suitable internships or job openings aligned with each student's profile.

This module could also allow companies to post opportunities directly on the platform, creating a bridge between learners and industry. Students would benefit from personalized job recommendations, application tracking, and career guidance, while instructors could align course content with current market demands. This would make the platform not only a learning system but also a career development hub.

7.2.2 Accessibility Enhancements

Improving accessibility is another important area for future development. The Skill Forge app can be enhanced by introducing features that make learning more inclusive for users with different needs. This could include options such as sign language support for video content, voice assistance for navigation, subtitles and transcripts for lectures, and customizable text sizes and color contrasts.

These accessibility features would ensure that students with disabilities can effectively use the platform without barriers. By providing multiple ways to interact with content, the app would create a more inclusive learning environment where every user can participate equally and comfortably.

7.2.3 Multi-Language Support:

To enhance accessibility and expand the reach of the Skill Forge app, multi-language support

can be implemented as a key future feature. This would allow users to select their preferred language for the entire application, including the user interface, course content, instructions, and subtitles for video lectures. By providing content in multiple languages, the platform would cater to a diverse group of learners from different linguistic backgrounds, making it easier for them to understand and engage with the material.

In addition, the system could include features such as real-time subtitle translation, language switching without restarting the course, and localized content recommendations based on the user's region and language preference. This enhancement would not only improve usability but also promote inclusivity, ensuring that language barriers do not limit access to quality education.

7.2.4 Enhanced Security

As the app grows and handles more sensitive data, including personal information, academic records, and payment details, it will be crucial to strengthen security measures. Future updates could include more robust encryption protocols for data storage and transmission, multi-factor authentication (MFA) for accessing accounts, and regular security audits to identify and address potential vulnerabilities.

Additionally, implementing advanced user access control features will ensure that only authorized users such as administrators, instructors, or students can access certain areas of the platform. The introduction of security protocols, such as biometric login (e.g., fingerprint or face recognition), would further enhance security, protecting user privacy and reducing the likelihood of unauthorized access.

7.2.5 Scalability

As more users join the platform, ensuring that the app can scale to meet the growing demand is paramount. Future improvements should focus on optimizing the app's backend architecture to support an increasing number of users, data transactions, and real-time interactions. This includes enhancing server capacity, optimizing database queries, and implementing content delivery networks (CDNs) to ensure fast and reliable content delivery globally.

Additionally, adopting cloud computing infrastructure can provide greater flexibility and scalability to meet varying levels of user demand, without compromising the app's performance or reliability. The ability to scale smoothly will ensure that the app remains responsive and accessible, even during peak usage periods.

7.2.6 Mobile App Enhancements

Expanding and enhancing the mobile version of the Skill Forge app for both Android and iOS

platforms will play a critical role in ensuring a seamless and consistent user experience across devices. Future updates should focus on improving the app's user interface (UI) and user experience (UX) to make navigation intuitive, fast, and efficient. Enhancing mobile features like push notifications, offline access, and in-app messaging will ensure that students and instructors can stay connected and engaged regardless of their location.

Further development could also include optimizing the app for tablets and larger screen devices to ensure the best experience for all users, providing them with a versatile and responsive learning platform.

7.2.7 Offline Access and Cloud Integration

For students who may not have consistent access to the internet, incorporating offline access to learning materials will be an invaluable addition. Students would be able to download course materials, lectures, assignments, and notes to their devices and access them offline, ensuring that their learning is uninterrupted, even in areas with limited connectivity.

Additionally, cloud integration will allow for seamless synchronization of data across multiple devices, ensuring that any progress made offline is automatically updated once the student has access to the internet again. This feature would provide greater flexibility for students, ensuring they can continue learning regardless of their network environment.

In summary, while the Skill Forge app is already a powerful and effective e learning tool, there is ample room for continued growth and enhancement. The future developments outlined above will ensure that the app remains competitive, secure, and user-centric, ultimately providing an even more engaging and efficient learning experience for all users. By focusing on real-time interactivity, personalized learning, enhanced security, and scalability, Skill Forge can continue to innovate and lead the way in modern educational technology.

Furthermore, this enhancement will improve accessibility by allowing users to learn at their own pace without being dependent on continuous internet connectivity. Efficient data caching mechanisms will ensure that downloaded content is stored securely and can be easily managed by the user. Cloud integration will also support automatic backup of user progress, reducing the risk of data loss. It will enable users to switch between devices without losing continuity in their learning experience. Additionally, synchronization processes will be optimized to minimize data usage and ensure quick updates. This feature will also support scalable storage solutions to handle increasing user data efficiently. Overall, it will strengthen the reliability and flexibility of the Skill Forge platform.

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